EFFICIENT COMPUTATION OF TRIPLE SCHUBERT STRUCTURE CONSTANTS

MATTHEW J. SAMUEL

Abstract. We present `schubmult`, an algorithm that expands a product of two double Schubert polynomials in different sets of variables into the double Schubert basis, and we bound its computational cost by $2^{f(n)+O(\log n)}$ coefficient-ring operations, where $f(n) = 2n\log_2 n + (3-2\log_2 e)n + \Theta(\log^2 n)$. For ordinary Schubert polynomials, the $\Theta(\log^2 n)$ term may be deleted. In the Grassmannian case (Schur and factorial Schur polynomials), the exponent collapses to $4(2n-k)H_2(k/(2n-k)) + \Theta(\log^2 n)$, with H_2 the binary entropy function. At the balanced dimension $k = n/2$, this is $2n\log_2(27/4) \approx 5.5098n$, yielding a cost of $(27/4)^{2n+o(n)}$ against $n^{\Theta(n)}$ in general; the maximum over k is $8n\log_2\phi \approx 5.5539n$ for ϕ the golden ratio, attained at $k \approx 0.553n$, so the balanced case is within one percent of the worst. The analysis rests on two structural facts: every permutation arising in the computation, output or intermediate, lies in a single interval of the left weak order, bounding their number by $(2n-1)!!$; and $\mathfrak{S}_v(x;z)$ admits a signed expansion into products of factorial elementary symmetric polynomials with variable counts λ_i for a set of admissible partitions λ, the counts being a free parameter. Because the product $\mathfrak{S}_u(x;y)\,\mathfrak{S}_v(x;z)$ has at least $|[e,v]_L|$ terms for every u, yielding $n!$ terms at $v = w_0(n)$, the cost for each fixed u is at most the cube of the worst-case output size, making `schubmult` worst-case output-polynomial. The algorithm is implemented in the Python package `schubmult`, available on the Python Package Index.

Keywords: Schubert polynomial, structure constant, Littlewood-Richardson rule, coproduct, double Schubert polynomial

Mathematics classification codes: 05E05, 05A05, 05A15, 14N10

Email: matthematics@gmail.com

1. Introduction

Multiplying two Schubert polynomials and expanding the result back in the Schubert basis is the central computational problem of Schubert calculus. The resulting structure constants c^w_{uv} are the intersection numbers of Schubert varieties in a flag manifold; in the Grassmannian case they specialize to the Littlewood–Richardson coefficients, and in the double (equivariant) case they become polynomials in two families of coefficient variables. This paper is about computing the entire expansion, and about proving what that computation costs.

1.1. The problem.

Almost all of what is known concerns a single coefficient. Computing one Littlewood–Richardson coefficient is $\#P$-complete [10]; deciding whether it is nonzero is, by contrast, in P [7, 4]. For Schubert polynomials the coefficients c^w_{uv} generalize the Littlewood–Richardson coefficients, so computing one is $\#P$-hard, while membership in $\#P$ is open, no combinatorial rule being known; the vanishing problem was recently placed in Σ^p_2 [11]. These results are discussed in Remark 5.22.

None of this tells us what it costs to produce the whole product, which is what an implementation actually does, and which is what a user of such a package actually wants. A complete expansion is not a single number but a whole table of them, and the natural accounting is not per output bit but per combinatorial witness. Such an accounting is only half of a complexity result, and the subject has stalled before at exactly the missing half. Remmel and Whitney [12] close their paper on multiplying Schur functions with a bound linear in the number of tableaux their algorithm produces, and then stop: the first step toward a worst-case or average-case statement, they observe, is to estimate that number, and they know of no work on the question. An output-sensitive bound becomes a complexity bound only when the size of the output is itself under control. We supply both halves here. The cost bound is Theorem 5.11; the missing half is Proposition 5.18, which bounds the number of terms below. It rests on Theorem 5.16, a vanishing criterion for the double structure constants which reduces every positive degree to the classical degree-0 case, and it needs from that case only the fact that a product of ordinary Schubert polynomials is nonzero—no estimate of the intractable degree-0 count enters (Remark 5.19). Our aim is a bound of that kind: explicit, provable, and sharp enough to expose exactly which structural feature of the double case is expensive.

1.2. **Main results.** Fix $u, v \in S_n$. The algorithm **schubmult**(u, v) of Method 5.1 computes the full expansion of $\mathfrak{S}_u(x; y)\, \mathfrak{S}_v(x; z)$ in the Schubert basis. Our results about it are the following.

A support bound of $(2n-1)!!$. Let $w_0 = w_0(n)$ be the longest element of S_n and let $w_1 = w_1(n)$ be the dominant permutation with $\mathfrak{c}(w_1) = \mathfrak{c}(w_0) + \mathfrak{c}(w_0) = (2n-2, 2n-4, \ldots, 2, 0)$. Every permutation appearing in the product, or at any intermediate stage of its computation, satisfies $w \leq_L w_1$ in the left weak order, and the number of such permutations is

$$(2n-1)!! \;=\; 1 \cdot 3 \cdot 5 \cdots (2n-1).$$

This is Lemma 5.3; the proof is a Cauchy-formula factorization of $\mathfrak{S}_{w_1}$ into two copies of $\mathfrak{S}_{w_0}$, followed by linear independence of the Schubert basis. The point is that the doubled staircase w_1, and not S_n or S_{2n-1}, is what confines the computation.

An explicit complexity bound. Theorem 5.11 bounds the number of coefficient-ring operations by

$$O\big(\, n^2\, \Phi_v \;+\; L \cdot P_u \cdot B \cdot n^3 \;+\; L \cdot P_u \cdot V \cdot B \cdot \Delta \cdot E_n \,\big)\,,$$

whose three summands are the one-time v-side precomputation, the live Pieri enumeration, and the matched-pair ring work. Written as $2^{\,f(n)+O(\log n)}$, this is

$$f(n) = 2n \log_2 n + (3 - 2\log_2 e)\, n + \Theta(\log^2 n),$$

so the worst case is bounded by $n^{\Theta(n)}$; when v is dominant this improves to $f(n) = n \log_2 n + (2 - \log_2 e)\, n + \Theta(\log^2 n)$. The leading term is exactly the support bound above: the $(2n-1)!!$ permutations on the u-side and the at most $n!$ labels on the v-side multiply to $(2n)!/2^n$, and Stirling's approximation converts this into the stated closed form.

A representation with free variable counts, and stable integer coefficients. Theorem 4.1 expands a double Schubert polynomial as a signed sum, over chains, of products of factorial elementary symmetric polynomials, and does so for any partition λ of variable counts subject only to the reachability condition $v \leq_L \mu_\lambda$, where μ_λ is the dominant permutation with $\mathfrak{c}(\mu_\lambda^{-1}) = \lambda$:

$$\mathfrak{S}_v(x; z) = \sum_{(v_0, \ldots, v_k)} \prod_{i=1}^{k} (-1)^{\sigma_i(v_{i-1}, v_i)}\, E_{\lambda_i - \ell(v_{i-1}, v_i),\ \lambda_i}\big(x; z_{D_i(v_{i-1}, v_i)}\big),$$

the chains running from the identity to $v\mu_\lambda^{-1}$. The counts are therefore a parameter of the representation rather than data extracted from v. A product of this kind is a standard elementary monomial when the counts are strictly decreasing, which is one admissible choice; the minimal choice $\lambda = \theta(v^{-1})$, which **elemrepr** uses because it minimizes the number of columns, is another. Corollary 4.2 shows that all of them carry the same integer data: setting $z = 0$ leaves the indexing set and the signs untouched, so the ordinary and double expansions are indexed by the same chains with the same integer coefficients, and one passes from the first to the second by the single substitution

$$e_p(x_1, \ldots, x_m) \;\longmapsto\; E_{p,m}\big(x; z_{D_i}\big),$$

replacing each elementary symmetric factor by the factorial elementary symmetric polynomial on the alphabet z_{D_i} that the chain already designates. Nothing combinatorial is added by the equivariant parameters; they only decorate factors that were going to be built anyway.

Where the cost of the double case sits. Corollary 5.13 gives the bound for ordinary Schubert polynomials, and it is that of Theorem 5.11 with the term $\Theta(\log^2 n)$ deleted. By the stability just described the two algorithms traverse identical witnesses with identical signs, so their entire difference is the weight attached to a witness, 1 in the ordinary case against one factorial elementary symmetric polynomial in the double case. Proposition 5.10 prices that weight exactly: building $E_{p,m}$ by the divide-and-conquer recursion costs $E_n = n^{\Theta(\log n)}$ ring operations, a quasi-polynomial factor, superpolynomial in n but subexponential. The bottleneck of the double case is thus coefficient-ring arithmetic rather than combinatorial explosion, which is why the implementation keeps every coefficient in factored form and never expands it into monomials.

The quantum case. Remark 4.4 describes the quantum setting, which fits the same framework with one localized exception adopted for speed. Worked in the SEM form throughout, with strictly decreasing counts, the quantum expansion is structurally identical to the classical one: it is a signed sum of products of factorial quantum elementary symmetric polynomials, indexed and signed as before, and the stability above applies verbatim. To go faster we consolidate certain pairs of factors into the double Schubert polynomial of the single permutation they jointly correspond to, replacing two Pieri steps by one; quantumly these consolidated factors can be kept as quantum double Schubert polynomials that behave correctly. Only they leave the elementary symmetric world, and only away from $q = 0$; every other factor is a factorial quantum elementary symmetric polynomial. The Pieri formulas the consolidated factors require are new, with proofs long enough to warrant separate papers, and we forward-cite them to [14, 13], in preparation. The architecture of **schubmult** is indifferent to this substitution.

The Grassmannian collapse, in n and k. When both factors are k-Grassmannian the product is Schur, respectively factorial Schur, multiplication. Lemma 6.2 confines every intermediate object to the $k \times 2(n-k)$ box, so both support counts and both per-step branchings drop to $G_k = \binom{2n-k}{k}$, and Theorem 6.3 and Corollary 6.5 reduce the exponent to

$$4(2n-k)\, H_2\!\left(\frac{k}{2n-k}\right), \qquad H_2(x) = -x\log_2 x - (1-x)\log_2(1-x).$$

The constant 4 in front is the substantive content, and a big-O reading would discard it. The case to hold in mind is the balanced one $k = n/2$, where $G_{n/2} = \binom{3n/2}{n/2} = (27/4)^{\,n/2+o(n)}$ and the exponent is $2n\log_2(27/4) = 5.5098\ldots n$, so the cost is $(729/16)^{\,n+o(n)}$. Nothing is lost by fixing k there: the maximum over all k is $8n\log_2\varphi \approx 5.554\,n$, with $\varphi = (1+\sqrt{5})/2$ the golden ratio, attained near $k \approx 0.553\,n$, less than one percent higher. Only the degenerate dimensions do better, the exponent shrinking to $\Theta(\log^2 n)$ as $k \to 0$ or $k \to n$, mirroring the shrinking of $\mathrm{Gr}(k,n)$ at the extremes. The running time is thus $2^{O(n)}$ for every k, where the general bound is $n^{\Theta(n)}$: a saving of a factor $n^{\Theta(n)}$. Read against the output rather than against n, the same bound says that the entire product is computed in time polynomial of degree at most four in G_k, which bounds the number of coefficients returned. Since $G_k = \Theta(n^k)$ for fixed k, this is also where the familiar statement that the count is polynomial in n comes from; Remark 5.22 explains why a bound of this shape, rather than polynomiality in the binary input size, is the most that can be asked here, and how far it sits from the information-theoretic floor. Read against the output rather than against n, the same bound says that the entire product is computed in time polynomial of degree at most four in G_k, which bounds the number of coefficients returned. Since $G_k = \Theta(n^k)$ for fixed k, this is also where the familiar statement that the count is polynomial in n comes from; Remark 5.22 explains why a bound of this shape, rather than polynomiality in the binary input size, is the most that can be asked here, and how far it sits from the information-theoretic floor.

1.3. **The algorithm.** The computation rests on the expansion of Theorem 2.3, which writes a double Schubert polynomial as a signed sum of products of factorial elementary symmetric polynomials indexed by chains, together with the Pieri formula of Theorem 2.4. It is assembled from three routines, each analyzed separately.

pieri (Section 3) enumerates the u-side covers of a single Pieri step. The enumeration is multiplicity-free (Lemma 3.3), and costs $O(n^3)$ per permutation produced (Proposition 3.4).

elemrepr (Section 4) builds the v-side, and is where Theorem 4.1 and its stability corollary are proved. A dominant double Schubert polynomial factors, one factorial elementary symmetric polynomial per column, and a general v is reached from a dominant one by divided differences applied column by column. Carried out naively this signed descending expansion produces large numbers of terms that eventually cancel. The dominant permutation itself supplies the filter that discards them: only endpoints beneath it in weak order are retained (Lemma 4.8), so no cancelling label ever seeds the next column and the number of live paths does not blow up.

schubmult (Section 5) yokes the two sides. It proceeds in layers, one per column of the shape, maintaining coefficients indexed by pairs (intermediate u-side permutation, v-side label). Two features drive the analysis. The v-side is precomputed once, so its transitions are table lookups; and within a layer the Pieri enumeration is run once per distinct u-side permutation rather than once per key, which decouples it from the number of v-side labels. This is what splits the cost into the three independent summands of Theorem 5.11.

1.4. **Implementation.** The three routines are not idealized descriptions of a computation carried out on paper. They are the routines of a program: **schubmult** is implemented in the Python package of the same name [16], distributed through the Python Package Index and installable in the usual way,

```
pip install schubmult
```

with sources and a command line interface included. The names **pieri**, **elemrepr** and **schubmult** are the names of the corresponding components there, so the complexity statements proved below are statements about code that exists and runs rather than about a design.

We record this for two reasons. The bounds of Theorem 5.11 count coefficient-ring operations, and a reader who wishes to see how that count behaves against wall-clock time, or on inputs beyond the reach of hand computation, can simply run the program. More importantly, every structural claim made here has a computational counterpart that the package makes checkable: the confinement of intermediates to a single weak order interval (Lemma 5.3), the coefficient stability of Corollary 4.2, the freedom in the counts λ afforded by Theorem 4.1, and the collapse of the Grassmannian case (Section 6) can each be verified directly on any given input. We have used it throughout in exactly that way.

1.5. **Organization.** Section 2 fixes notation and records the two formulas the algorithm is built on. Sections 3, 4 and 5 present **pieri**, **elemrepr** and **schubmult** with their correctness and cost statements, and the complexity analysis is assembled in Theorem 5.11 and Corollary 5.13, with prior work discussed in Remark 5.22. Section 6 treats the Grassmannian case. The implementation is described in Section 1.4.

2. Foundational formulas

2.1. Notation and definitions.

Definition 2.1 ($\mathfrak{c}$, dominant permutations, θ, μ)**.** We say that a permutation μ is *dominant* if $\mathfrak{c}_i(\mu) \geq \mathfrak{c}_{i+1}(\mu)$ for all i. For a permutation $v \in S_\infty$, we define a dominant permutation $\mu(v)$ recursively as follows. If v is already dominant, define $\mu(v) = v$. Otherwise, let i be the maximal index such that $\mathfrak{c}_i(v) < \mathfrak{c}_{i+1}(v)$, and define

$$\mu(v) = \mu(vs_i)$$

Define $\theta_i(v)$ to be $\mathfrak{c}_i(\mu(v))$.

Definition 2.2 ($\overset{i}{\Rightarrow}$, $\sigma_i(u,v)$, $D_i(u,v)$)**.** Let $u, v \in S_\infty$ and $i \geq 1$ be an integer. We define $u \overset{i}{\Rightarrow} v$ if there exist distinct integers $b_1, b_2, \ldots, b_k$ such that $b_j \neq i$ for all j, $\ell(ut_{i,b_1} \cdots t_{i,b_j}) = \ell(u) + j$ for all j, and

$$v = ut_{i,b_1} \cdots t_{i,b_k}$$

We define

$$\sigma_i(u,v) = \#\{j \mid b_j < i\}$$

and we define

$$D_i(u,v) = \{u(i)\} \cup \{u(j) \mid u(j) \neq v(j)\}$$

2.2. The formula for double Schubert polynomials.

Theorem 2.3 (Formula for double Schubert polynomials)**.** *Suppose $v \in S_\infty$. Then*

$$\mathfrak{S}_v(x;z) = \sum_{(v_0,\ldots,v_k)} \prod_{i=1}^{k} (-1)^{\sigma_i(v_{i-1},v_i)} E_{\theta_i(v)-\ell(v_{i-1},v_i),\theta_i(v)}(x; z_{D_i(v_{i-1},v_i)})$$

Where $v_{i-1} \overset{i}{\Rightarrow} v_i$ for all i, $v_0 = 1$, and $v_k = v\mu(v^{-1})$.

The counts $\theta_i(v)$ appearing here are the minimal admissible ones. Theorem 4.1 below shows that the formula persists with $\theta(v)$ replaced by any partition λ that reaches v, the endpoint changing accordingly; the statement above is the case $\lambda = \theta(v^{-1})$.

2.3. The Pieri formula.

Theorem 2.4 (The Pieri formula, [15, Theorem 7.1])**.** *Suppose $u \in S_\infty$, $k \geq 1$, and $0 \leq p \leq k$. Let x, y, z be sets of variables. Then*

$$\mathfrak{S}_u(x;y)\, E_{p,k}(x;z) = \sum_{u \xrightarrow{k} w} E_{p-\ell(u,w),k-\ell(u,w)}(y; z_{P_k(u,w)})\, \mathfrak{S}_w(x;y)$$

where $P_k(u,w) = \{u(i) \mid i \leq k \text{ and } u(i) = w(i)\}$.

3. The **pieri** algorithm

Definition 3.1 (k-strip cover, admissible k-chain, the (p,k)-Pieri problem)**.** Let $w \in S_n$, let $1 \leq k < n$, and let $\beta \in \{1,\dots,n\} \cup \{\infty\}$. A *$k$-strip cover of* (w,β) is a transposition t_{ab} of positions a,b with $1 \leq a \leq k < b \leq n$ such that

(1) $w \lessdot wt_{ab}$ in Bruhat order; that is, $w(a) < w(b)$ and no c with $a < c < b$ satisfies $w(a) < w(c) < w(b)$; and
(2) $w(b) < \beta$.

Its *output* is the pair $(wt_{ab},\, w(b))$. An *admissible k-chain of length p* from u is a sequence $u = w_0, w_1, \dots, w_p$ carrying bounds $\infty = \beta_0 > \beta_1 > \cdots > \beta_p$ in which (w_t, β_t) is the output of a k-strip cover of (w_{t-1}, β_{t-1}) for every t. The *(p,k)-Pieri problem* for u asks to enumerate every admissible k-chain from u of length at most p. These chains index the permutations occurring in $e_p(x_1,\dots,x_k)\,\mathfrak{S}_u$, where e_p is the elementary symmetric polynomial.

Method 3.2 (**pieri**(u,p,k))**.** Given $u \in S_n$, $1 \leq k < n$, and $0 \leq p \leq k$, the algorithm returns a list T of pairs (w,t), one for each admissible k-chain from u of length $t \leq p$, recording its endpoint w.

(1) Initialize $T \leftarrow [(u,0)]$ and a frontier $F \leftarrow [(u,\infty)]$.
(2) For $t = 1,\dots,p$, form a new frontier $F' \leftarrow [\,]$ and, for each $(w,\beta) \in F$:
   (a) compute $A \leftarrow \{\, a \in \{1,\dots,k\} \mid w(a) < \beta \,\}$;
   (b) for each $b = k+1,\dots,n$ with $w(b) < \beta$, and each $a \in A$ for which t_{ab} is a k-strip cover of (w,β), append $(wt_{ab},\, w(b))$ to F' and $(wt_{ab},\, t)$ to T.

   Then set $F \leftarrow F'$.
(3) Return T.

Lemma 3.3 (Multiplicity-freeness)**.** *For each t, distinct admissible k-chains from u of length t have distinct endpoints; equivalently, the endpoint map is a bijection from admissible k-chains of length t onto the permutations w with $w \gtrdot^k u$ of length $\ell(u)+t$ produced by the Pieri rule for $e_t(x_1,\dots,x_k)$.*

Proof. The strictly decreasing bounds $\beta_0 > \beta_1 > \cdots$ force the values moved leftward across position k to be strictly decreasing, which is exactly the condition making the e_t-Pieri expansion multiplicity-free; see [2]. Reconstructing the chain from its endpoint by peeling off these moves in decreasing order of value shows the map is injective, hence bijective onto the Pieri support. ☐

Proposition 3.4 (Complexity of **pieri**)**.** *Let $M := |T|$ be the number of admissible k-chains from u of length at most p that* **pieri**(u,p,k) *returns. The algorithm runs in*

$$O(k\,(n-k)\,n \cdot M)$$

elementary operations, that is $O(n^3)$ per enumerated permutation. Moreover, by Lemma 3.3,

$$M \;\leq\; \#\{\, w \in S_n \mid w \geq u,\ \ell(w) \leq \ell(u) + p \,\}.$$

Proof. Processing one frontier node (w,β) costs $O(k)$ to build A, and then iterates over the $n-k$ choices of b and the at most k elements of A; for each pair the Bruhat-cover test and the transposition each scan $O(n)$ positions. Thus a node costs $O\big(k + (n-k)\,k\,n\big) = O\big(k\,(n-k)\,n\big)$. By Lemma 3.3 the number of nodes ever placed on a frontier equals $\sum_{t=0}^{p-1} N_t$, where N_t is the number of length-t chains; since $\sum_{t=0}^{p-1} N_t \leq \sum_{t=0}^{p} N_t = M$, the total cost is $O\big(k\,(n-k)\,n \cdot M\big)$. The bound on M holds because every endpoint satisfies $w \geq u$ and $\ell(w) \leq \ell(u) + p$. ☐

4. The **elemrepr** algorithm

We now describe an algorithm **elemrepr** for expressing a double Schubert polynomial as in Theorem 2.3. Its starting point is the structure of a dominant double Schubert polynomial. If μ is dominant with $\mathfrak{c}(\mu) = (\lambda_1,\dots,\lambda_m)$, then $\mathfrak{S}_\mu(x;z)$ factors as a product of factorial elementary symmetric polynomials of maximal degree, one per column of the shape,

$$\mathfrak{S}_\mu(x;z) = \prod_{i=1}^{m} E_{\lambda_i,\,\lambda_i}(x;z_i),$$

where z_i is the indicated single indeterminate [8]. For a general v one writes $\mathfrak{S}_v(x;z)$ by applying divided differences in the z-variables to the dominant polynomial $\mathfrak{S}_{\mu(v^{-1})}(x;z)$. By the Leibniz formula, this operation may be carried out one column at a time. Applied to a single column this produces a signed descending expansion.

What **elemrepr** produces is thus a representation of $\mathfrak{S}_v$ as a signed sum of products of elementary symmetric polynomials in nested initial alphabets. A product of this kind is a *standard elementary monomial* (SEM) when its variable counts are strictly decreasing. The representation used here does not require that, and in fact the counts are not determined by v at all: any partition large enough to reach v may serve. The following theorem is the general statement, of which Theorem 2.3 is the minimal case.

Theorem 4.1 (Elementary symmetric representation of a double Schubert polynomial)**.** *Let $v \in S_\infty$ and let λ be a partition. Let μ_λ denote the dominant permutation determined by*

$$\mathfrak{c}(\mu_\lambda^{-1}) = \lambda,$$

and suppose that $v \leq_L \mu_\lambda$ in the left weak order. Then

$$\mathfrak{S}_v(x;z) = \sum_{(v_0,\ldots,v_k)} \prod_{i=1}^{k} (-1)^{\sigma_i(v_{i-1},v_i)}\, E_{\lambda_i-\ell(v_{i-1},v_i),\ \lambda_i}\big(x; z_{D_i(v_{i-1},v_i)}\big),$$

the sum being over all chains $1 = v_0 \overset{1}{\Rightarrow} v_1 \overset{2}{\Rightarrow} \cdots \overset{k}{\Rightarrow} v_k$ with endpoint

$$v_k = v\,\mu_\lambda^{-1}.$$

Proof. Since μ_λ is dominant, so is μ_λ^{-1}, and the dominant double Schubert polynomial factors by columns,

$$\mathfrak{S}_{\mu_\lambda^{-1}}(x;z) = \prod_i E_{\lambda_i,\,\lambda_i}(x;z_i),$$

one maximal-degree factorial elementary symmetric polynomial per part of $\lambda = \mathfrak{c}(\mu_\lambda^{-1})$. The hypothesis $v \leq_L \mu_\lambda$ says exactly that $\ell(v\mu_\lambda^{-1}) = \ell(\mu_\lambda) - \ell(v)$, so $\mathfrak{S}_v(x;z)$ is reached from this dominant polynomial by a reduced sequence of $\ell(\mu_\lambda) - \ell(v)$ divided differences in the z-variables, the passage from μ_λ to v being by left multiplication by length-shortening generators. By the Leibniz formula these operators act one column at a time, and on the single column i the effect is the signed descending expansion described above: a step $v_{i-1} \overset{i}{\Rightarrow} v_i$ lowers the degree of the factor E_{λ_i,λ_i} to $E_{\lambda_i-\ell(v_{i-1},v_i),\,\lambda_i}$, contributes the sign $(-1)^{\sigma_i(v_{i-1},v_i)}$, and selects the coefficient alphabet $z_{D_i(v_{i-1},v_i)}$. Composing over the columns and collecting the chains with endpoint $v\mu_\lambda^{-1}$ gives the stated sum. $\square$

The choice of λ is genuinely free, subject only to $v \leq_L \mu_\lambda$. The minimal admissible choice is $\lambda = \theta(v^{-1})$, for which $\mu_\lambda^{-1} = \mu(v^{-1})$ and the endpoint is $v\,\mu(v^{-1})$; the theorem then reduces to Theorem 2.3, and this is the choice **elemrepr** makes, since it minimizes the number of columns and hence the number of layers. Taking λ strictly decreasing recovers the SEM form. All such choices are available, and the next corollary applies to each of them.

Corollary 4.2 (The double upgrade fixes the integer coefficients)**.** *In the notation of Theorem 4.1, setting $z = 0$ gives*

$$\mathfrak{S}_v(x) = \sum_{(v_0,\ldots,v_k)} \prod_{i=1}^{k} (-1)^{\sigma_i(v_{i-1},v_i)}\, e_{\lambda_i-\ell(v_{i-1},v_i)}(x_1,\ldots,x_{\lambda_i}),$$

the sum being over the same chains as in that theorem. The two expansions are indexed by the identical set of chains and attach to each chain the identical integer coefficient $\prod_i (-1)^{\sigma_i(v_{i-1},v_i)}$; the double polynomial $\mathfrak{S}_v(x;z)$ is obtained from the ordinary one $\mathfrak{S}_v(x)$ by replacing each factor

$$e_p(x_1,\ldots,x_m) \;\longmapsto\; E_{p,m}\big(x; z_{D_i(v_{i-1},v_i)}\big),$$

that is, by substituting into it the alphabet of coefficient variables indexed by $D_i(v_{i-1},v_i)$.

Proof. Directly from the definition

$$E_{p,m}(x;z) = \sum_{1\le i_1<\cdots<i_p\le m} \prod_{j=1}^{p} (x_{i_j} - z_{i_j-j+1})$$

we have $E_{p,m}(x;0) = e_p(x_1,\ldots,x_m)$. Specializing $z=0$ in Theorem 4.1 therefore replaces each factor $E_{\lambda_i-\ell(v_{i-1},v_i),\lambda_i}(x; z_{D_i(v_{i-1},v_i)})$ by $e_{\lambda_i-\ell(v_{i-1},v_i)}(x_1,\ldots,x_{\lambda_i})$ and leaves the indexing set and the signs untouched. Since $\mathfrak{S}_v(x;0) = \mathfrak{S}_v(x)$, the displayed expansion follows, and reading the specialization backwards gives the stated substitution rule. □

The consequence for computation is that the z-alphabets are inert combinatorial data: they are selected by the chain, not produced by it. The whole equivariant content of the double case sits inside the substituted factors $E_{p,m}$, which is why **schubmult** and its single-variable specialization traverse identical witnesses with identical signs (Corollary 5.13), and why the only cost separating them is that of building one factorial elementary symmetric polynomial (Proposition 5.10).

Remark 4.3 (Stability is independent of the choice of counts)*.* Theorem 4.1 makes the counts a parameter: any partition λ with $v \le_L \mu_\lambda$ furnishes a representation, and Corollary 4.2 applies to each of them, since its proof uses nothing about λ beyond the identity $E_{p,m}(x;0) = e_p(x_1,\ldots,x_m)$, valid for every pair $p \le m$. The stability is therefore a property of the shape of the representation rather than of any one choice of counts. Whenever $\mathfrak{S}_v(x;z)$ is written as a signed sum, over some indexing set, of products of factorial elementary symmetric polynomials $E_{p_i,m_i}(x;z_{D_i})$ with integer coefficients, setting $z=0$ yields the expansion of $\mathfrak{S}_v(x)$ over the same indexing set, with the same integer coefficients, in the products of $e_{p_i}(x_1,\ldots,x_{m_i})$; and the double polynomial is recovered from the ordinary one by substituting the designated alphabets z_{D_i} back into the factors. The freedom in the counts is thus genuine rather than a defect: the strictly decreasing SEM counts are admissible, as is the minimal choice $\theta(v^{-1})$ used here, as is every partition between them, and all of them carry the same integer data.

Remark 4.4 (The quantum case)*.* The quantum theory fits the framework of Remark 4.3 with one localized exception, introduced deliberately and only for speed.

Taking the counts strictly decreasing, that is working in the SEM form throughout, everything above carries over: the expansion is a signed sum of products of factorial quantum elementary symmetric polynomials, indexed and signed exactly as in the classical case, and the stability of Corollary 4.2 applies verbatim with those factors in place of the classical ones. No structural obstruction arises. The cost of this uniformity is speed, since the strict counts forgo the savings available from repeated parts.

The optimization recovers those savings. In the classical expansion certain pairs of factorial elementary symmetric factors are consolidated into the double Schubert polynomial of the single permutation they jointly correspond to, replacing two Pieri steps by one. Passing to the quantum setting, in specific cases these consolidated factors may be retained as quantum double Schubert polynomials, which continue to behave correctly under the algorithm. Only these consolidated factors leave the elementary symmetric world: they are not elementary symmetric polynomials unless one specializes $q=0$, at which point the whole representation reverts to the classical elementary symmetric one. Every unconsolidated factor remains a factorial quantum elementary symmetric polynomial. The almost strict medium shape is what records which pairs have been consolidated.

The Pieri formulas required for the consolidated factors are new and are not proved here; their proofs are substantial enough to require separate treatment, and we forward-cite them to [14] and [13], in preparation. The algorithmic architecture of this paper is unaffected: the layered structure, the one-time v-side precomputation, and the decoupling of the Pieri enumeration from the label count all carry over unchanged, with these formulas substituted for their classical counterparts where a consolidation has been made.

Definition 4.5 (k-strip descent beneath a bound, the down-Pieri problem)**.** Let $w, \mu \in S_n$ and $1 \le k < n$. Say w' lies *beneath the bound* μ if

$$\mathrm{inv}(w'\mu) = \mathrm{inv}(\mu) - \mathrm{inv}(w'),$$

that is, $\ell(w'\mu) = \ell(\mu) - \ell(w')$. A *$k$-strip descent* of w is a Bruhat descent $w \gtrdot wt_{ab}$ realized by a transposition t_{ab}, with $a<b$ and $k \in \{a,b\}$, that moves the value in the pivot position k either leftward (if $b=k$, with sign -1) or rightward (if $a=k$, with sign $+1$), touching only positions still holding their original value. Given a starting label w, the permutation μ, a degree p, and the column k, the *down-Pieri problem* asks for the

signed list of those length-t descending k-strip chains from w ($t \le p$) whose endpoint lies beneath μ in weak order.

Method 4.6 (**kdown**(w, μ, p, k))**.** The algorithm returns a list D of triples (w', t, s), one for each length-t descending k-strip chain from w ($t \le p$) whose endpoint w' lies beneath μ in weak order, with $s \in \{+1, -1\}$ its sign.

(1) Initialize $D \leftarrow [\,]$; if w is already beneath μ, append $(w, 0, +1)$. Set a frontier $F \leftarrow [(w, +1)]$.
(2) For $t = 1, \dots, p$, form $F' \leftarrow [\,]$ and, for each $(w'', s) \in F$:
   (a) (leftward) for each position $a < k$ with $w''(a) = w(a)$ such that $w'' \gtrdot w'' t_{a,k}$ is a Bruhat descent, set $w''' \leftarrow w'' t_{a,k}$ and append $(w''', -s)$ to F'; if w''' is beneath μ, append $(w''', t, -s)$ to D;
   (b) (rightward) for each position $b > k$ with $w''(b) = w(b)$ such that $w'' \gtrdot w'' t_{k,b}$ is a Bruhat descent, set $w''' \leftarrow w'' t_{k,b}$ and append $(w''', +s)$ to F'; if w''' is beneath μ, append $(w''', t, +s)$ to D.

   Then set $F \leftarrow F'$.
(3) Return D.

Applied without the bound μ, the signed descending expansion produces many extraneous terms that ultimately cancel. The dominant permutation μ is exactly the bound that discards them: only endpoints beneath μ are retained in D. In the assembled v-side (Method 4.7 below) only these retained labels seed the next column, so the bound prevents the number of live paths from blowing up across columns.

Method 4.7 (**elemrepr**(v), the v-side path dictionaries)**.** Let $\theta := \theta(v^{-1})$ with nonzero parts in columns $1, \dots, L$, and let $vmu := v\,\mu(v^{-1})$. Processing columns from $i = L$ down to $i = 1$, maintain for each column a set of live labels (initialized at column L to the single label vmu). At column i, with $k = i$ and bound $\mu_i := \mu$ of the shape truncated below column i, call **kdown**(w, μ_i, θ_i, k) for each live label w; record its signed output triples as the transitions out of w, and let their endpoints be the live labels seeding column $i - 1$. The collected transitions form the path dictionaries used by **schubmult**.

Lemma 4.8 (Correctness of the bound filter)**.** *A permutation w' produced by a descending k-strip chain from w is retained by* **kdown**(w, μ, p, k) *if and only if w' lies beneath μ, i.e. $\operatorname{inv}(w'\mu) = \operatorname{inv}(\mu) - \operatorname{inv}(w')$. In the assembly of Method 4.7 the retained signed labels are exactly the terms of $\mathfrak{S}_v(x;z)$ contributed through column i, so no extraneous (cancelling) label propagates to column $i - 1$.*

Proof. The retention test is the identity $\operatorname{inv}(w'\mu) = \operatorname{inv}(\mu) - \operatorname{inv}(w')$ checked in step (2); this is precisely the condition that $w' \le \mu$ in weak order, i.e. $\ell(w'\mu) = \ell(\mu) - \ell(w')$. The factorization of the dominant polynomial by column, together with the column-wise decoupling of the z-divided differences, identifies the beneath-μ endpoints at column i with the surviving monomials of $\mathfrak{S}_v(x;z)$ after processing that column; the extraneous chains are exactly those leaving the shape, which the bound discards before they seed column $i - 1$. $\square$

Proposition 4.9 (Complexity of **kdown**/**elemrepr**)**.** *Let $R := |D|$ be the number of retained signed triples and F^* the total number of frontier nodes generated by* **kdown**(w, μ, p, k)*. The algorithm runs in*

$$O\big(n^2 \cdot F^*\big)$$

elementary operations, and $R \le F^*$. Consequently **elemrepr**(v)*, which invokes* **kdown** *once per live label across the $L \le \min(n-1, \ell_v)$ columns and by Lemma 4.8 propagates only retained labels, runs in $O\big(n^2 \cdot \Phi_v\big)$ operations, where Φ_v is the total number of frontier nodes generated over all columns.*

Proof. Processing one frontier node scans $O(n)$ candidate positions, each requiring an $O(n)$ Bruhat-descent test and $O(n)$ transposition, so $O(n^2)$ per node; summing over the F^* nodes gives $O(n^2 F^*)$. Each retained triple arises from a distinct frontier node, so $R \le F^*$. In **elemrepr**, by Lemma 4.8 the labels seeding column $i - 1$ are only the retained (beneath-bound) endpoints of column i, so the live-label count never includes extraneous paths; summing the per-column frontier work gives $O(n^2 \Phi_v)$. $\square$

5. The **schubmult** algorithm

Method 5.1 (The algorithm **schubmult**)**.** Fix n and let $u, v \in S_n$. The algorithm **schubmult**(u, v) computes the expansion of $\mathfrak{S}_u(x;y)\,\mathfrak{S}_v(x;z)$ in the Schubert basis as follows. Set $\theta := \theta(v^{-1})$ and let

$$L := \#\{i \mid \theta_i \ne 0\}$$

be the number of nonzero parts.

The arithmetic takes place in the coefficient ring $R := \mathbb{Z}[y; z]$, in which the structure constants live. The algorithm proceeds in L layers and maintains a hash H that maps each pair (w, λ)—an intermediate u-side permutation w together with a v-side label λ—to a coefficient $c(w, \lambda) \in R$. All coefficients are initialized to 0 except for the single key (u, e), which is initialized to $c(u, e) = 1$. Write $\mu := \mu(v^{-1})$ and $vmu := v\mu$.

Each layer transforms H by one column of the shape. Fix layer i and put $k := \theta_i$, and group the keys of H by their u-side permutation. For each distinct u-side permutation w appearing in some key:

(1) Call **pieri**(w, θ_i, θ_i) (Method 3.2) once to enumerate the u-side covers $w \xrightarrow{\theta_i} w'$; each such cover carries a degree $a \in \{0, 1, \ldots, \theta_i\}$, which is number of inversions it adds. The resulting cover list is reused for every label paired with w and is not recomputed per label.
(2) For each label λ with $c = c(w, \lambda) \neq 0$, read from **elemrepr**(v) (Method 4.7) the v-side transitions out of λ in column i: each is a signed triple (λ', b, s) with endpoint λ', degree b, and sign $s \in \{+1, -1\}$.
(3) For every pair with $w \xrightarrow{\theta_i} w'$ of degree a, triple $(\lambda', b, s))$ with $a + b \leq \theta_i$, accumulate into the new hash the single ring element
$$c(w', \lambda') \ \mathrel{+}= \ s \cdot c \cdot E_{\theta_i - a - b,\ \theta_i - a}\big(y_A\,;\, z_B\big),$$
where $E_{p,m}(y; z)$ denotes the factorial elementary symmetric polynomial of degree p in the first m variables,
$$E_{p,m}(y; z) := \sum_{1 \leq i_1 < \cdots < i_p \leq m} \prod_{j=1}^{p} \big(y_{i_j} - z_{\,i_j - j + 1}\big),$$
evaluated on the y-alphabet $y_A := \{\, y_{w'(j)} \mid j \leq \theta_i,\ w(j) = w'(j) \,\}$ of the $\theta_i - a$ values the cover leaves fixed in columns $1, \ldots, \theta_i$, and the z-alphabet z_B where
$$B = \{i\} \cup \{\lambda(j) \mid j \neq i \text{ and } \lambda(j) = \lambda'(j)\}$$
(4) After all L layers the coefficient $c(w,\, vmu)$ at each u-side endpoint w is the structure constant of $\mathfrak{S}_w(x; y)$ in $\mathfrak{S}_u(x; y)\ \mathfrak{S}_v(x; z)$.

Thus each layer performs, for every matched (cover, triple) pair, the construction of one factorial elementary symmetric polynomial $E_{p,m}(y; z)$, followed by one ring multiplication (by the sign and the running coefficient) and one ring addition into the accumulator $c(w', \lambda')$. The polynomial $E_{p,m}(y; z)$ is rebuilt on each pair by the divide-and-conquer recursion of Lemma 5.2: this recursion is not memoized in the y-alphabet—only the z-index selection is cached—so its construction is a nontrivial cost, recorded as E_n in Definition 5.8, and not a shared lookup. Every coefficient $c(w, \lambda)$ is kept in factored form—a product of the factorial elementary symmetric factors $E_{p,m}(y; z)$ accumulated along its paths—and is never multiplied out into the monomial basis, so it asymptotically stays far smaller than its monomial expansion, whose number of terms is exponential in n.

Lemma 5.2 (Divide-and-conquer splitting of $E_{p,m}$)**.** *Let $y = (y_1, \ldots, y_m)$ and let $z = (z_r)_{r \in \mathbb{Z}}$ be an alphabet. For an integer c put*
$$E^{(c)}_{p,m}(y; z) := \sum_{1 \leq i_1 < \cdots < i_p \leq m} \prod_{j=1}^{p} \big(y_{i_j} - z_{\,i_j - j + 1 + c}\big),$$
so that $E_{p,m} = E^{(0)}_{p,m}$. Set $t := \lfloor m/2 \rfloor$, $y' := (y_1, \ldots, y_t)$, and $y'' := (y_{t+1}, \ldots, y_m)$. Then for $0 \leq p \leq m$ and every integer c,
$$E^{(c)}_{p,m}(y; z) = \sum_{a=0}^{p} E^{(c)}_{a,t}(y'; z)\, E^{(c+t-a)}_{p-a,\, m-t}(y''; z),$$
with the conventions $E^{(c)}_{0,\ell} = 1$ and $E^{(c)}_{a,\ell} = 0$ for $a > \ell$, and the base cases $E^{(c)}_{1,\ell}(y; z) = \sum_{i=1}^{\ell}(y_i - z_{i+c})$ and $E^{(c)}_{\ell,\ell}(y; z) = \prod_{i=1}^{\ell}(y_i - z_{1+c})$.

Proof. Classify each index set $1 \leq i_1 < \cdots < i_p \leq m$ in the defining sum of $E^{(c)}_{p,m}$ by the number a of its members lying in $\{1, \ldots, t\}$; the remaining $p - a$ lie in $\{t+1, \ldots, m\}$. The first a factors $\prod_{j=1}^{a}\big(y_{i_j} - z_{i_j - j + 1 + c}\big)$

range over all size-a subsets of $\{1,\dots,t\}$ and sum to $E^{(c)}_{a,t}(y';z)$. For the upper part write $i_{a+r}=t+l_r$ with $1\le l_1<\cdots<l_{p-a}\le m-t$; the corresponding factor is

$$y_{t+l_r}-z_{\,i_{a+r}-(a+r)+1+c}=y''_{l_r}-z_{\,l_r-r+1+(c+t-a)},$$

which is the r-th factor of $E^{(c+t-a)}_{p-a,\,m-t}(y'';z)$. So the upper factors sum to $E^{(c+t-a)}_{p-a,\,m-t}(y'';z)$; the terms with $a>t$ contribute 0, consistent with the convention. Summing over a gives the identity. ☐

Lemma 5.3 (Output support bound of $(2n-1)!!$)**.** *Let $w_0=w_0(n)$ be the longest element of S_n, with $\mathfrak{c}(w_0)=(n-1,n-2,\dots,1,0)$, and let $w_1=w_1(n)$ be the permutation with*

$$\mathfrak{c}(w_1)=\mathfrak{c}(w_0)+\mathfrak{c}(w_0)=(2n-2,\,2n-4,\,\dots,\,2,\,0),$$

a dominant element of S_{2n-1} with $\ell(w_1)=2\binom{n}{2}$. For any $u,v\in S_n$, every permutation w occurring in the product $\mathfrak{S}_u\cdot\mathfrak{S}_v$, or in any intermediate product formed during its computation, satisfies $w\le_L w_1$. The number of such permutations is at most $(2n-1)!!$.

Proof. Consider that

$$\mathfrak{S}_{w_1(n)}(x;y_1,z_1,\dots,y_{n-1},z_{n-1})=\mathfrak{S}_{w_0(n)}(x;y_1,\dots,y_{n-1})\,\mathfrak{S}_{w_0(n)}(x;z_1,\dots,z_{n-1}).$$

Expanding via the Cauchy formula, we obtain

$$\sum_{w\le_L w_1(n)}\mathfrak{S}_w(x)\,\mathfrak{S}_{ww_1(n)^{-1}}(-y_1,-z_1,\dots,-y_{n-1},-z_{n-1})=\sum_{u,v}\mathfrak{S}_u(x)\,\mathfrak{S}_v(x)\,\mathfrak{S}_{uw_0(n)}(-y)\,\mathfrak{S}_{vw_0(n)}(-z)$$

By linear independence of the Schubert basis, every w occurring in the product $\mathfrak{S}_u(x)\,\mathfrak{S}_v(x)$ with nonzero coefficient must satisfy $w\le_L w_1(n)$. The number of permutations w with $w\le_L w_1(n)$ is the size of the left weak order interval $[e,w_1(n)]_L$, the double factorial

$$(2n-1)!!=1\cdot3\cdot5\cdots(2n-1)=\frac{(2n-1)!}{2^{\,n-1}\,(n-1)!}.$$

This counts all outputs, intermediate ones included. ☐

The bound $(2n-1)!!$ is a worst-case statement, obtained by replacing both factors by w_0. For a given pair u,v the relevant dominant permutation is smaller, and the interval beneath it can be counted exactly rather than estimated. We record both refinements.

Lemma 5.4 (Exact size of a dominant weak order ideal)**.** *Let $\sigma\in S_N$ be dominant and write $\lambda=\mathfrak{c}(\sigma)$. Then*

$$\bigl|[e,\sigma]_R\bigr|=\frac{N!}{\prod_{i=1}^{N}\bigl(\sigma(i)-\lambda_i\bigr)},$$

and consequently, for dominant μ,

$$\bigl|[e,\mu]_L\bigr|=\bigl|[e,\mu^{-1}]_R\bigr|=\frac{N!}{\prod_{i=1}^{N}\bigl(\mu^{-1}(i)-\mathfrak{c}_i(\mu^{-1})\bigr)},$$

the inverse of a dominant permutation being again dominant. Both expressions are unchanged by padding σ with fixed points, so N may be taken to be any value at least the support size.

Proof. In the right weak order $w\le_R\sigma$ if and only if $\mathrm{Inv}(w)\subseteq\mathrm{Inv}(\sigma)$, where $\mathrm{Inv}(w)=\{(i,j)\mid i<j,\ w(i)>w(j)\}$. Order $\{1,\dots,N\}$ by the non-inversions of σ, setting $i\prec j$ when $i<j$ and $\sigma(i)<\sigma(j)$; this relation is transitive, and the condition $\mathrm{Inv}(w)\subseteq\mathrm{Inv}(\sigma)$ says exactly that $i\prec j$ implies $w(i)<w(j)$. Hence $|[e,\sigma]_R|$ is the number of linear extensions of the poset P_σ.

We claim P_σ is a rooted forest, that is, the elements above any fixed j form a chain. If not, there are $k<k'$ both above j and mutually incomparable, so $j<k<k'$ with $\sigma(j)<\sigma(k')<\sigma(k)$, an occurrence of the pattern 132; but a permutation is dominant precisely when it is 132-avoiding. By Knuth's hook length formula for forests the number of linear extensions is $N!/\prod_i|{\downarrow}i|$, where ${\downarrow}i=\{j\mid j\preceq i\}$. Since $\prec$ is transitive, ${\downarrow}i=\{j\le i\mid\sigma(j)\le\sigma(i)\}$, of size $1+\#\{j<i\mid\sigma(j)<\sigma(i)\}$. Finally

$$\sigma(i)-1=\#\{j<i\mid\sigma(j)<\sigma(i)\}+\#\{j>i\mid\sigma(j)<\sigma(i)\}=\#\{j<i\mid\sigma(j)<\sigma(i)\}+\lambda_i,$$

so $|{\downarrow}i|=\sigma(i)-\lambda_i\ge1$, giving the first display. The second follows because $w\mapsto w^{-1}$ is an isomorphism from $[e,\mu]_L$ onto $[e,\mu^{-1}]_R$. Adjoining a fixed point at position $N+1$ contributes the single new factor $N+1$ and replaces $N!$ by $(N+1)!$, leaving the quotient unchanged. ☐

Lemma 5.5 (Instance form of the support bound)**.** *For $u \in S_n$ let μ_u denote the dominant permutation determined by $\mu_u^{-1} = \mu(u^{-1})$, so that $u \leq_L \mu_u$, and define μ_v likewise. Let M be the dominant permutation with*

$$\mathfrak{c}(M) = \mathfrak{c}(\mu_u) + \mathfrak{c}(\mu_v).$$

Then every permutation w occurring in $\mathfrak{S}_u \cdot \mathfrak{S}_v$, or in any intermediate product formed during its computation, satisfies $w \leq_L M$, and the number of such permutations is $|[e, M]_L|$, evaluated exactly by Lemma 5.4.

Proof. Identical to the proof of Lemma 5.3, with $\mathfrak{S}_{\mu_u}$ and $\mathfrak{S}_{\mu_v}$ in place of the two copies of $\mathfrak{S}_{w_0}$: the dominant polynomial $\mathfrak{S}_M$ in the doubled alphabet factors as $\mathfrak{S}_{\mu_u}\mathfrak{S}_{\mu_v}$ because the codes add, and expanding both sides by the Cauchy formula and comparing coefficients in the Schubert basis confines the support to $[e, M]_L$. Since $u \leq_L \mu_u$ and $v \leq_L \mu_v$, the pair (u, v) is among those so expanded. For $u = v = w_0$ we have $\mu(w_0) = w_0$, hence $\mu_u = \mu_v = w_0$ and $\mathfrak{c}(M) = \mathfrak{c}(w_0) + \mathfrak{c}(w_0) = \mathfrak{c}(w_1)$. □

Remark 5.6 (What the refinement does and does not buy). Lemma 5.5 is an exact count of an interval, not an estimate, and for most pairs it is far smaller than $(2n-1)!!$. It is nonetheless still an overcount of the quantity actually needed: the permutations arising in the computation satisfy $w \geq u$ in Bruhat order and $\ell(w) \leq \ell_u + \ell_v$, and $|[e, M]_L|$ imposes neither restriction, counting the whole ideal below M. Nor does it improve the worst-case bound, since $u = v = w_0$ gives $\mu_u = \mu_v = w_0$ and $M = w_1$, returning the count $(2n-1)!!$ exactly and recovering Lemma 5.3. The refinement is therefore instance-adaptive: it sharpens the bound for any particular input while leaving the exponent of Theorem 5.11 unchanged. A useful special case, immediate from Lemma 5.4 applied to a strictly decreasing code, is the elementary bound

$$\big|[e, M]_L\big| \;\leq\; \prod_i \big(\mathfrak{c}_i(M) + 1\big),$$

with equality exactly when the nonzero parts of $\mathfrak{c}(M)$ are strictly decreasing, as they are for $M = w_1$.

Corollary 5.7 (Closed form for the instance support count)**.** *For $u, v \in S_n$ let $M = M(u, v)$ be as in Lemma 5.5 and write*

$$\Lambda(u, v) := \big|[e, M]_L\big| = \frac{N!}{\prod_{i=1}^{N}\big(M^{-1}(i) - \mathfrak{c}_i(M^{-1})\big)}, \qquad N := 2n - 1.$$

Then $\Lambda(u, v) \leq (2n-1)!!$ for all u, v, with equality at $u = v = w_0$, and $\Lambda(e, e) = 1$. When one factor is w_0,

$$\Lambda(w_0, v) = \prod_{i=1}^{n}\big(n - i + 1 + \mathfrak{c}_i(\mu_v)\big),$$

which decreases from $(2n-1)!!$ at $v = w_0$ to $n!$ at $v = e$.

Proof. Everything is Lemma 5.4 applied to M, together with Lemma 5.5 for $\Lambda(w_0, w_0) = |[e, w_1]_L| = (2n-1)!!$. When one factor is w_0 the code $\mathfrak{c}(w_0) + \mathfrak{c}(\mu_v)$ is strictly decreasing, consecutive parts differing by $1 + \mathfrak{c}_i(\mu_v) - \mathfrak{c}_{i+1}(\mu_v) \geq 1$, so the product form of Remark 5.6 is attained. At $v = w_0$ we have $\mathfrak{c}(\mu_v) = (n-1, \ldots, 1, 0)$ and the product is $\prod_i (2(n-i)+1) = (2n-1)!!$; at $v = e$ it is $\prod_i (n-i+1) = n!$. Padding M with fixed points leaves the quotient unchanged, so any N at least the support size may be used. □

For two general factors $\mathfrak{c}(\mu_u) + \mathfrak{c}(\mu_v)$ is only weakly decreasing, so the product form is a strict overcount; the hook expression remains exact in all cases.

Definition 5.8 (ℓ_u, ℓ_v, P_u, V, B, Δ, E_n)**.** For $u, v \in S_n$ write $\ell_u := \ell(u)$ and $\ell_v := \ell(v)$, and set

$$P_u := \#\{\, w \mid u \leq w \leq_L M \text{ and } \ell(w) \leq \ell_u + \ell_v \,\},$$

with $M = M(u, v)$ as in Lemma 5.5, so that every intermediate and output u-side permutation is counted by P_u, and

$$P_u \;\leq\; \big|[e, M]_L\big| \;\leq\; (2n-1)!!.$$

Write $\theta := \theta(v^{-1})$, so that $\mu_v^{-1} = \mu(v^{-1})$ is the dominant permutation with $\mathfrak{c}(\mu_v^{-1}) = \theta$ (Lemma 5.5), and let V be the number of distinct v-side labels produced by **elemrepr**(v) across all columns, so that

$$V \;\leq\; \big|[e, v\mu_v^{-1}]\big| \;\leq\; n!,$$

where an unsubscripted interval denotes Bruhat order. The two *per-layer branchings* are

$$B := \max_i \max_w \left|\mathbf{pieri}(w, \theta_i, \theta_i)\right|, \qquad \Delta := \max_i \max_\lambda \#\{\text{kdown triples } (\lambda', c, \varepsilon) \text{ carried by } \lambda \text{ in column } i\} :$$

B is the maximum number of u-side covers generated by a single **pieri** step (Proposition 3.4), and Δ is the maximum number of v-side kdown triples generated from a single label (Proposition 4.9). Each branching enumerates a family of k-strips indexed by (nonempty) subsets of at most n eligible positions, so

$$B \leq 2^n - 1, \qquad \Delta \leq 2^n - 1.$$

Finally, let $R(p, m)$ denote the number of coefficient-ring operations (additions and multiplications) performed by the unmemoized divide-and-conquer recursion of Lemma 5.2 in evaluating $E^{(c)}_{p,m}(y; z)$; the count is independent of the offset c. With $t := \lfloor m/2 \rfloor$ it satisfies $R(p, m) = 0$ for $p = 0$ or $p > m$, the base values $R(1, \ell) = R(\ell, \ell) = \ell - 1$, and, for $1 < p < m$,

$$R(p, m) = R(p, t) + R(p, m - t) + 1 + \sum_{a=\max(1,\, p-m+t)}^{\min(p,\, t)} \big(R(a, t) + R(p - a,\, m - t) + 2\big).$$

Define

$$E_n := \max_{0 \leq p \leq m \leq n-1} R(p, m).$$

Remark 5.9. The middle quantity in $P_u \leq \big|[e, M]_L\big| \leq (2n - 1)!!$ is evaluated exactly by Lemma 5.4, and the outer inequality is attained at $u = v = w_0$. Every v-side label is reached from the initial label $v\mu_v^{-1}$ by a sequence of descending k-strip transpositions, each of which is a Bruhat descent, so all labels lie weakly below $v\mu_v^{-1}$, whence $V \leq \big|[e, v\mu_v^{-1}]\big|$. In the resulting chain $V \leq \big|[e, v\mu_v^{-1}]\big| \leq n!$ the first inequality is the instance-adaptive form and the second the crude bound of Lemma 5.3. The second is usually far from tight, since $\ell(v\mu_v^{-1}) = \ell(\mu_v) - \ell(v)$ collapses as $\ell(v)$ approaches $\ell(\mu_v)$, vanishing outright when v is dominant. The bound $B \leq 2^n - 1$ is sharp, attained at $w = w_0$, where $\mathbf{pieri}(w_0, n - 1, n)$ produces exactly $2^n - 1$ covers.

Proposition 5.10. $\log_2 E_n = \Theta(\log^2 n)$*; equivalently* $E_n = 2^{\Theta(\log^2 n)} = n^{\Theta(\log n)}$.

Proof. Write $C(m) := \max_{0 \leq p \leq m} R(p, m)$, so $E_n = \max_{m \leq n-1} C(m)$. Throughout put $t := \lfloor m/2 \rfloor$, so the two child sizes t and $m - t$ lie in $\{\lfloor m/2 \rfloor, \lceil m/2 \rceil\}$, and let N be the number of admissible indices a in the recurrence, so $N \leq t + 1 \leq m$.

Upper bound. For $1 < p < m$ the recurrence has $2N + 2$ summands of the form $R(\cdot, t)$ or $R(\cdot, m - t)$, each at most $C(\lceil m/2 \rceil)$, together with $2N + 1$ additive constants; hence, using $C(\lceil m/2 \rceil) \geq 1$ for $m \geq 3$,

$$R(p, m) \leq (2N + 2)\, C(\lceil m/2 \rceil) + (2N + 1) \leq (4m + 3)\, C(\lceil m/2 \rceil).$$

Since $C(2) = 1$, induction on m gives $C(m) \leq (4m + 3)^{\lceil \log_2 m \rceil}$: the step uses $4\lceil m/2 \rceil + 3 \leq 4m + 3$ and $1 + \lceil \log_2 \lceil m/2 \rceil \rceil = \lceil \log_2 m \rceil$. Therefore

$$\log_2 C(m) \leq \lceil \log_2 m \rceil \, \log_2(4m + 3) = O(\log^2 m),$$

and so $\log_2 E_n = O(\log^2 n)$.

Lower bound. Let $I(p, m)$ count the internal nodes—calls with $1 < p < m$—of the unmemoized recursion tree of $R(p, m)$. Each internal node performs at least one ring addition, so $R(p, m) \geq I(p, m)$, and I satisfies, for $1 < p < m$,

$$I(p, m) = 1 + I(p, t) + I(p, m - t) + \sum_{a=\max(1,\, p-m+t)}^{\min(p,\, t)} \big(I(a, t) + I(p - a,\, m - t)\big).$$

Put $J(m) := \sum_{p=0}^{m} I(p, m)$. Dropping all terms of the sum except $I(a, t)$, a fixed value $a^* \in \{1, \ldots, t\}$ is an admissible index in the recurrence for $R(p, m)$ precisely when $a^* \leq p \leq a^* + (m - t)$, which holds for at least $\lfloor m/2 \rfloor$ values of p with $1 < p < m$; summing the resulting contribution $I(a^*, t)$ over those p gives

$$J(m) \;\geq\; \left\lfloor \tfrac{m}{2} \right\rfloor \sum_{a^*=1}^{t} I(a^*, t) \;=\; \left\lfloor \tfrac{m}{2} \right\rfloor J(t).$$

As $J(4) \geq 1$ (since $I(2,4) \geq 1$), unrolling this inequality down to a fixed constant size yields

$$\log_2 J(m) \;\geq\; \sum_{i=0}^{\lfloor \log_2 m \rfloor - 3} \log_2 \left\lfloor \tfrac{m}{2^{\,i+1}} \right\rfloor \;=\; \Omega(\log^2 m).$$

Finally $J(m) \leq (m+1)\max_p I(p,m) \leq (m+1)\,C(m)$, so $\log_2 C(m) \geq \log_2 J(m) - \log_2(m+1) = \Omega(\log^2 m)$, whence $\log_2 E_n \geq \log_2 C(n-1) = \Omega(\log^2 n)$. ☐

Theorem 5.11 (Complexity of **schubmult**)**.** *Write $L := \min(n-1, \ell_v)$. The number of coefficient-ring operations performed by* **schubmult**(u,v) *is*

$$O\big(\, n^2\,\Phi_v \;+\; L \cdot P_u \cdot B \cdot n^3 \;+\; L \cdot P_u \cdot V \cdot B \cdot \Delta \cdot E_n \,\big)\,,$$

where P_u, V, B, Δ, E_n are as in Definition 5.8 and Φ_v is the v-side precomputation cost of Proposition 4.9. Writing that bound as $2^{\,f(n)+O(\log n)}$,

$$f(n) = 2n\log_2 n + (3 - 2\log_2 e)\,n + \Theta(\log^2 n),$$

and, when v is dominant, $f(n) = n\log_2 n + (2 - \log_2 e)\,n + \Theta(\log^2 n)$, where $\log_2 e = 1/\ln 2 \approx 1.4427$.

Remark 5.12. The three summands of Theorem 5.11 are, respectively, the one-time v-side precomputation, the live **pieri** enumeration (run once per distinct u-side permutation per layer), and the matched-pair ring work; the third dominates in the worst case. Here B and Δ are the u-side (Pieri) and v-side (kdown-triple) per-layer branchings and E_n is the per-iteration elementary-symmetric construction cost, superpolynomial in n but subexponential. Each term of $f(n)$ is a multiplicative factor of the running time, so all are retained. The dominant contribution is the product of the two support counts: the at most

$$P_u \leq (2n-1)!! = \frac{(2n-1)!}{2^{\,n-1}\,(n-1)!}$$

distinct u-side permutations and the at most $V \leq n!$ distinct v-side labels, whose product telescopes,

$$P_u V \;\leq\; (2n-1)!!\; n! \;=\; \frac{(2n)!}{2^{\,n}},$$

so together with the branching factor $B\Delta \leq 4^n$ the two counts and branchings contribute $\log_2(P_u V) + 2n = \log_2((2n)!) + n$; the elementary-symmetric construction adds $\log_2 E_n = \Theta(\log^2 n)$. Applying Stirling in the form $\log_2(m!) = m\log_2 m - m\log_2 e + O(\log m)$ gives $\log_2((2n)!) = 2n\log_2 n + 2n - 2n\log_2 e + O(\log n)$, so

$$f(n) = \log_2((2n)!) + n + \Theta(\log^2 n) = 2n\log_2 n + (3 - 2\log_2 e)\,n + \Theta(\log^2 n),$$

and the worst-case running time is bounded by $2^{\,f(n)+O(\log n)} = n^{\Theta(n)} = 2^{\Theta(n\log n)}$. The live **pieri** enumeration contributes only n^3 per distinct u-permutation and so lands in the $O(\log n)$ term; it is decoupled from the v-label count V because **pieri**(w,θ_i,θ_i) is run once per permutation w present in a layer, independent of λ.

Proof. By Method 5.1 the algorithm runs in $L = \#\{i \mid \theta_i(v^{-1}) \neq 0\} \leq \min(n-1, \ell_v)$ layers. Write $\mu := \mu(v^{-1})$, so that $\sum_i \theta_i(v^{-1}) = \ell(\mu)$; note that $\ell(\mu) \geq \ell_v$, with equality if and only if v is dominant. The two sides are yoked: at layer i a u-side degree u_i and a v-side degree v_i are paired with $u_i + v_i = \theta_i$, and the v-side descends from $v\mu$ to the identity, so $\sum_i v_i = \ell(v\mu)$. Hence the total degree added on the u-side is

$$\sum_i u_i = \sum_i \theta_i - \sum_i v_i = \ell(\mu) - \ell(v\mu) = \ell_v,$$

using $\ell(v\mu) = \ell(\mu) - \ell(v)$ (as $v^{-1} \leq \mu$ in weak order). Thus every intermediate u-side permutation satisfies $w \geq u$ and $\ell(w) \leq \ell_u + \ell_v$; by Lemma 5.5 it also satisfies $w \leq_L M$, hence $w \leq_L w_1(n)$ and $w \in S_{2n-1}$, so there are at most P_u of them. The v-side labels are permutations obtained from $v\mu$ by descending k-strip transpositions; each such transposition is a Bruhat descent, so every label lies weakly below $v\mu$ in Bruhat order and by Definition 5.8 there are at most $V \leq \big|[e, v\mu]\big| \leq n!$ distinct ones, $\mu = \mu_v^{-1}$. The hash at the start of each layer therefore has at most $P_u \cdot V$ keys.

We account for the work in three parts. First, the entire v-side is precomputed once by **elemrepr**(v) (Method 4.7) at cost $O(n^2\Phi_v)$ ring operations (Proposition 4.9); thereafter each v-side transition is a table lookup carrying no combinatorial work.

Second, the live **pieri** enumeration. Because every u-side cover strictly raises length, the u-side transitions form a graded acyclic graph, so within a fixed layer no permutation recurs; hence $\mathbf{pieri}(w, \theta_i, \theta_i)$ is invoked at most once per distinct u-side permutation present in that layer (Method 5.1, step 1). The number of such invocations is

$$\sum_{i=1}^{L} \left|\Phi_i^u\right| \;\leq\; L \cdot P_u,$$

where Φ_i^u is the set of distinct u-side permutations live at layer i; when consecutive layers share a degree, $\theta_i = \theta_{i+1}$, caching **pieri** on (w, θ_i) collapses this to at most P_u invocations, and the safe bound $L \cdot P_u$ costs only the extra factor $L \leq n$. Each invocation costs $O(n^3)$ per enumerated cover (Proposition 3.4) and returns at most B covers, so the total **pieri** cost is $O(L \cdot P_u \cdot B \cdot n^3)$. This enumeration is run once per permutation, not once per key, so it is independent of the v-label count V.

Third, the matched-pair ring work. Write Φ_i for the keys (w, λ) present at layer i. The inner loop matches one **pieri** cover of w against one kdown triple of λ, executed

$$W \;=\; \sum_{i=1}^{L} \sum_{(w,\lambda)\in\Phi_i} \beta_i(w)\, \delta_i(\lambda)$$

times, where $\beta_i(w) := |\mathbf{pieri}(w, \theta_i, \theta_i)|$ and $\delta_i(\lambda)$ is the number of kdown triples carried by λ in column i. Bounding $\beta_i(w) \leq B$, $\delta_i(\lambda) \leq \Delta$, and $|\Phi_i| \leq P_u \cdot V$ gives $W \leq L \cdot P_u \cdot V \cdot B \cdot \Delta$. Each executed iteration constructs one factorial elementary symmetric polynomial $E_{p,m}(y; z)$ (with $p \leq m \leq \theta_i \leq n-1$) by the unmemoized divide-and-conquer recursion of Lemma 5.2, a fresh evaluation costing E_n ring operations (Definition 5.8); the ensuing sign multiplication, ring addition, and $O(n)$ bookkeeping are lower-order. Hence the matched-pair cost is $O(W \cdot E_n) = O(L \cdot P_u \cdot V \cdot B \cdot \Delta \cdot E_n)$.

Summing the three parts gives

$$T = O\big(n^2 \Phi_v + L \cdot P_u \cdot B \cdot n^3 + L \cdot P_u \cdot V \cdot B \cdot \Delta \cdot E_n\big)\,,$$

the bound of Theorem 5.11. Since $\Phi_v \leq L \cdot V \cdot (2^n - 1)$—each of the at most $L \cdot V$ kdown calls generates at most $2^{\theta_i} - 1 \leq 2^n - 1$ frontier nodes—and $V \cdot \Delta \cdot E_n \geq n^3$ in the worst case, the third summand dominates there.

For the worst case take base-2 logarithms of the third summand and substitute $L \leq n$, $P_u \leq (2n-1)!!$ (Lemma 5.3), $V \leq n!$, $B, \Delta \leq 2^n - 1$, and $\log_2 E_n = \Theta(\log^2 n)$ (Proposition 5.10), using $(2n-1)!!\; n! = (2n)!/2^{\,n}$ to combine the two support counts with the $+2n$ from $B\Delta$:

$$\log_2 T \;\leq\; \log_2\big((2n)!\big) + n + \Theta(\log^2 n) + O(\log n),$$

which is $f(n) + O(\log n)$. We retain every summand: the two support counts and the branchings contribute $\log_2((2n)!) + n$, and the elementary-symmetric construction contributes $\log_2 E_n = \Theta(\log^2 n)$; only the live **pieri** cost, at $3\log_2 n$, falls into the $O(\log n)$ slack. By Stirling $\log_2((2n)!) = 2n\log_2 n + 2n - 2n\log_2 e + O(\log n)$, so $f(n) = 2n\log_2 n + (3 - 2\log_2 e)\, n + \Theta(\log^2 n)$, giving $T = n^{\Theta(n)} = 2^{\Theta(n \log n)}$. If v is dominant then $\theta(v) = \mathfrak{c}(v)$, $\ell(\mu) = \ell_v$, every path step carries a $+1$ sign, and each layer's v-side is a single path, so $\Delta = 1$ and only $O(\ell_v)$ labels are live in place of $V = n!$; then $T = O(L \cdot P_u \cdot \ell_v \cdot B \cdot E_n)$ and $\log_2 T \leq \log_2((2n-1)!!) + n + \Theta(\log^2 n) + O(\log n)$. Since $(2n-1)!! = (2n)!/(2^n n!)$, Stirling gives $\log_2((2n-1)!!) = n\log_2 n + (1 - \log_2 e)\, n + O(\log n)$, so in this case $f(n) = n\log_2 n + (2 - \log_2 e)\, n + \Theta(\log^2 n)$. $\square$

Corollary 5.13 (Ordinary Schubert polynomials)**.** *For ordinary (single) Schubert polynomials, computing the structure constants of the product $\mathfrak{S}_u(x)\; \mathfrak{S}_v(x)$ by the single-variable specialization of* **schubmult** *takes*

$$O\big(\, n^2\, \Phi_v \;+\; L \cdot P_u \cdot B \cdot n^3 \;+\; L \cdot P_u \cdot V \cdot B \cdot \Delta \,\big)$$

integer operations, with $L = \min(n-1, \ell_v)$. Writing the bound as $2^{\,g(n)+O(\log n)}$, the worst case is bounded by

$$g(n) = 2n\log_2 n + (3 - 2\log_2 e)\, n,$$

and, when v is dominant, $g(n) = n\log_2 n + (2 - \log_2 e)\, n$.

Proof. The single-variable algorithm has the same combinatorial skeleton as Method 5.1: the shape $\theta := \theta(v^{-1})$, the u-side Pieri covers, and the v-side path structure with its signed transitions (λ', b, s) are identical, so the three-part accounting of the proof of Theorem 5.11—the one-time v-side precomputation $O(n^2\Phi_v)$, the live **pieri** enumeration $O(L \cdot P_u \cdot B \cdot n^3)$ run once per distinct u-side permutation, and the matched-pair work over $W \le L \cdot P_u \cdot V \cdot B \cdot \Delta$ pairs—carries over verbatim. The single difference is the weight attached to a matched (cover, triple) pair: for double Schubert polynomials it is the factorial elementary symmetric polynomial $E_{p,m}(y;z)$, whereas for ordinary Schubert polynomials the corresponding Pieri coefficient is 1, so the pair contributes only the integer sign $s \in \{+1,-1\}$. Each executed iteration therefore performs one integer multiplication by s and one integer addition—$O(1)$ ring operations—so E_n is replaced by $O(1)$ in the third summand. Taking base-2 logarithms of the dominant (third) summand and substituting $L \le n$, $P_u \le (2n-1)!!$ (Lemma 5.3), $V \le n!$, $B, \Delta \le 2^n - 1$ (and $\Delta = 1$ in the dominant case), and combining $(2n-1)!!\, n! = (2n)!/2^n$ with the $+2n$ from $B\Delta$, gives $g(n)$, which is $f(n)$ of Theorem 5.11 without the $\Theta(\log^2 n)$ contribution of E_n. □

Corollary 5.14 (Instance-adaptive exponent)**.** *For any $u, v \in S_n$ the bound of Theorem 5.11 may be written $2^{f(u,v)+O(\log n)}$ with*

$$f(u,v) = \log_2 \Lambda(u,v) + \log_2\bigl|[e, v\mu_v^{-1}]\bigr| + 2n + \Theta(\log^2 n),$$

where $\Lambda(u,v)$ is the closed form of Corollary 5.7 and the second interval is taken in Bruhat order. Taking $u = v = w_0$ gives $\log_2 \Lambda = \log_2((2n-1)!!)$ and $\bigl|[e, v\mu_v^{-1}]\bigr| \le n!$, recovering $f(n)$ of Theorem 5.11, since $(2n-1)!!\, n! = (2n)!/2^n$. Taking $u = w_0$ and v arbitrary gives the v-adaptive exponent

$$f(w_0, v) = \sum_{i=1}^{n} \log_2\bigl(n - i + 1 + \mathfrak{c}_i(\mu_v)\bigr) + \log_2\bigl|[e, v\mu_v^{-1}]\bigr| + 2n + \Theta(\log^2 n).$$

Proof. In the proof of Theorem 5.11 the dominant summand contributes $\log_2 T \le \log_2 L + \log_2 P_u + \log_2 V + \log_2 B + \log_2 \Delta + \log_2 E_n + O(1)$. Substitute $P_u \le \Lambda(u,v)$ and $V \le \bigl|[e, v\mu_v^{-1}]\bigr|$ (Definition 5.8), $B, \Delta \le 2^n - 1$, $L \le n$, and $\log_2 E_n = \Theta(\log^2 n)$ (Proposition 5.10); the branchings contribute $2n$ and L falls into the $O(\log n)$ term. The specializations are those of Corollary 5.7 together with $v\mu_v^{-1} \in S_n$, and $\log_2((2n-1)!!) - \log_2(n!) = n \log_2 n + O(n)$ by Stirling. □

Remark 5.15 (Which term collapses). Both of the first two terms of $f(u,v)$ are instance quantities: the first counts the u-side permutations, the second the v-side labels. The first alone ranges over an interval of width $\log_2\bigl((2n-1)!!/n!\bigr) = n \log_2 n + O(n)$ as μ_v runs from e to w_0, but the bottom of that range is vacuous: $\mu_v = e$ forces $v = e$, and there is nothing to compute. The only genuine collapse is in the second term, which vanishes exactly when v is dominant. Then $\mu_v = v$, the initial label $v\mu_v^{-1}$ is the identity, so $V = 1$ and **elemrepr** returns the column factorization of $\mathfrak{S}_v(x;z)$ with no descending expansion at all. A dominant v may be as long as w_0, leaving the first term at its maximum, so this is a real saving of $\log_2(n!) = n \log_2 n + O(n)$ from the worst-case exponent rather than a degenerate one.

Theorem 5.16 (Double support as a union of ordinary supports)**.** *For $u, v \in S_n$ write $c^w_{uv}(y;z)$ for the coefficients in*

$$\mathfrak{S}_u(x;y)\ \mathfrak{S}_v(x;z) = \sum_w c^w_{uv}(y;z)\ \mathfrak{S}_w(x;y),$$

and c^w_{uv} for the ordinary structure constants of $\mathfrak{S}_u(x)\ \mathfrak{S}_v(x)$. Then

$$c^w_{uv}(y;z) \ne 0 \qquad \Longleftrightarrow \qquad c^w_{uv''} \ne 0 \ \ \text{for some } v'' \le_L v \text{ with } \ell(u) + \ell(v'') = \ell(w),$$

equivalently

$$\operatorname{supp}\bigl(\mathfrak{S}_u(x;y)\ \mathfrak{S}_v(x;z)\bigr) \ = \ \bigcup_{v'' \le_L v} \operatorname{supp}\bigl(\mathfrak{S}_u(x)\ \mathfrak{S}_{v''}(x)\bigr).$$

Proof. Fix u and w. We may write

$$c^w_{u,w_0(n)}(x;z) = \sum_v \partial^w_u(\mathfrak{S}_v(x))\, \mathfrak{S}_{vw_0}(-z).$$

If $\ell(u) + \ell(v') = \ell(w)$, then we have the term $c^w_{u,v'}(0;0)\, \mathfrak{S}_{v'w_0(n)}(-z)$ in the sum. We also have that for all v with $vw_0(n) \le_L v'w_0(n)$, or equivalently $v' \le_L v$, that

$$\partial^{vw_0(n)}_{(z)}(c^w_{u,w_0(n)}(x;z)) = c^w_{u,v}(y;z).$$

It follows that if $c^w_{u,v'}(0;0) \ne 0$, then $c^w_{u,v}(y;z) \ne 0$ for all $v \ge_L v'$. This proves the direction $\Leftarrow$.

For the converse, suppose for some v that $c^w_{u,v'}(y;z) \neq 0$ for all $v' \geq_L v$. It follows that not only do we have that $c^w_{u,v}(0;z) \neq 0$, but also that $\deg_z(c^w_{u,v}(0;z)) = \ell(v)$. In particular, since some Schubert polynomial $\mathfrak{S}_{\widetilde{v}}(-z)$ with $\ell(\widetilde{v}) = \ell(v)$ has a nonzero coefficient, there exists an i such that

$$\partial^i_{(z)}(c^w_{u,v}(0;z)) = -c^w_{u,s_iv}(0;z) \neq 0$$

whence $c^w_{u,s_iv}(y;z) \neq 0$ and $\deg_z(c^w_{u,s_iv}(0;z)) = \ell(v) - 1$. Continuing inductively, it follows that there exists some $v'' \leq_L v$ with $\ell(u) + \ell(v'') = \ell(w)$ such that $c^w_{u,v''}(y;z) = c^w_{u,v''}(0;0) \neq 0$. This proves the direction $\Rightarrow$, and we are done. □

Remark 5.17 (Reduction of the double vanishing problem). The length condition in Theorem 5.16 determines the rank of v'': only the elements of $[e,v]_L$ of length exactly $\ell(w) - \ell(u)$ can witness w. In particular the double support meets every degree d with $\ell_u \leq d \leq \ell_u + \ell_v$, since $[e,v]_L$ is graded by length and a product of ordinary Schubert polynomials never vanishes.

The two directions of Theorem 5.16 behave quite differently. The union is over the left weak order ideal $[e,v]_L$, so no cancellation between its members can occur: the ordinary structure constants are nonnegative integers, and a permutation lying in the support of $\mathfrak{S}_u\,\mathfrak{S}_{v''}$ for even one $v'' \leq_L v$ already forces $c^w_{uv}(y;z) \neq 0$. The content of the theorem is the converse, that the polynomial $c^w_{uv}(y;z)$ cannot be nonzero for any other reason. Consequently, deciding whether a double structure constant vanishes is a disjunction of instances of the ordinary vanishing problem, the problem placed in Σ^p_2 by [11]; no separate theory of vanishing for the double constants is needed. The length condition bounds that disjunction by the number of elements of $[e,v]_L$ of length $\ell(w) - \ell(u)$, a single rank of the ideal rather than all $|[e,v]_L| \leq n!$ of it.

The degree bookkeeping is what makes the union spread. Assigning degree 1 to each of x, y, and z, the polynomial $\mathfrak{S}_u(x;y)$ is homogeneous of degree $\ell(u)$, so the coefficient $c^w_{uv}(y;z)$ is homogeneous of degree $\ell(u) + \ell(v) - \ell(w)$ in y and z. A permutation w with $\ell(w) < \ell(u) + \ell(v)$ is thus admitted not by any failure of homogeneity but by its coefficient carrying positive degree in the coefficient variables—which is exactly what the ordinary constants, being scalars, cannot do. This is why the ordinary support is confined to the single degree $\ell(u)+\ell(v)$ while the double support is not: each $v'' \leq_L v$ contributes in degree $\ell_u + \ell(v'')$ alone, and the decomposition of the double support by degree is precisely the decomposition of $[e,v]_L$ by rank,

Proposition 5.18 (Output lower bound, uniform in u)**.** *For all $u, v \in S_n$,*

$$\bigl|\operatorname{supp}\bigl(\mathfrak{S}_u(x;y)\ \mathfrak{S}_v(x;z)\bigr)\bigr| \ \geq\ \bigl|[e,v]_L\bigr|,$$

with equality at $u = e$. In particular, taking $v = w_0(n)$, the product $\mathfrak{S}_u(x;y)\ \mathfrak{S}_{w_0}(x;z)$ has at least $n!$ terms for every $u \in S_n$.

Proof. Work in $\mathbb{Q}(y,z)[x]$. The ordinary Schubert polynomials $\mathfrak{S}_{v''}(x)$ form a $\mathbb{Q}$-basis of $\mathbb{Q}[x]$, hence a $\mathbb{Q}(y,z)$-basis after extension of scalars, so they are linearly independent over $\mathbb{Q}(y,z)$. Since $\mathbb{Q}(y,z)[x]$ is an integral domain and $\mathfrak{S}_u \neq 0$, multiplication by $\mathfrak{S}_u$ is an injective $\mathbb{Q}(y,z)$-linear map, and therefore the $\bigl|[e,v]_L\bigr|$ polynomials

$$\bigl\{\ \mathfrak{S}_u(x)\ \mathfrak{S}_{v''}(x)\ :\ v'' \leq_L v\ \bigr\}$$

are linearly independent. By Theorem 5.16 every one of them is supported, in the Schubert basis, inside $\operatorname{supp}(\mathfrak{S}_u(x;y)\,\mathfrak{S}_v(x;z))$; so that set indexes a spanning family for a space of dimension at least $\bigl|[e,v]_L\bigr|$, and being itself linearly independent it has at least that many elements. Equality at $u = e$ is the case $\mathfrak{S}_u\,\mathfrak{S}_{v''} = \mathfrak{S}_{v''}$ of Theorem 5.16. For $v = w_0$ the ideal $[e,w_0]_L$ is all of S_n. □

Remark 5.19 (What the lower bound needs from degree 0). The difficulty this proposition addresses was stated plainly forty years ago. Remmel and Whitney [12] end their paper on multiplying Schur functions with a complexity discussion that arrives at an output-sensitive bound and halts there. Their algorithm traverses a tree whose leaves are the tableaux $C(\lambda * \mu)$ enumerating the product, at a cost of $K|\mu|\,\bigl|C(\lambda*\mu)\bigr|$ steps; having noted that the constant might be improved, they add that this "is not the main problem to determine the complexity of this algorithm," since

> the first step for either a worst case or average complexity result for this algorithm is to find some reasonable estimates for $|C(\lambda * \mu)|$. However, we do not know of any work done on estimates for $|C(\lambda * \mu)|$.

The obstruction is not incidental to their method. An output-sensitive bound is a complexity bound only in the presence of an estimate for the output, and estimating the output means determining how many of the coefficients vanish.

Theorem 5.16 is what makes that tractable here, and it does so as a vanishing criterion rather than as a way around one. Sort the terms of the double product by the degree $\ell(u)+\ell(v)-\ell(w)$ of the coefficient $c^w_{uv}(y;z)$, which is determined by $\ell(w)$. A witness v'' contributes in degree $\ell(v)-\ell(v'')$, so each w has a single degree attached to it, and degree 0 comes from the single witness $v''=v$: the coefficients of degree 0 are the ordinary structure constants c^w_{uv}, $\#P$-hard to compute and with vanishing known only to lie in Σ^p_2 [11]. The theorem gives necessary and sufficient conditions for $c^w_{uv}(y;z)\neq 0$, and thereby reduces vanishing in every positive degree to vanishing in degree 0, the case that has actually been studied (Remark 5.17). The positive-degree coefficients are not a separate difficulty; they are governed by the classical one.

What Proposition 5.18 draws from degree 0 is the least that could be drawn: not which ordinary constants vanish, nor how many, but only that in each ordinary product $\mathfrak{S}_u\,\mathfrak{S}_{v''}$ at least one does not. That much is free, $\mathbb{Q}[x]$ being a domain. Theorem 5.16 then propagates a single nonvanishing degree-0 coefficient of $\mathfrak{S}_u\,\mathfrak{S}_{v''}$ into a nonvanishing coefficient of the double product in degree $\ell(v)-\ell(v'')$, for every $v''\leq_L v$ at once, and the $\big|[e,v]_L\big|$ witnesses so obtained are what the bound counts. The proof is phrased through linear independence over $\mathbb{Q}(y,z)$ only because distinct witnesses of equal length contribute in the same degree, so that counting one term apiece requires an argument; the mechanism is the propagation.

Proposition 5.18 therefore counts no degree individually, and in particular never counts the degree-0 terms, which are what Remmel and Whitney were facing, the Grassmannian case being Littlewood–Richardson. The entire bound is carried by the terms of positive degree, and the case $u=e$ shows how completely. There the ordinary product $\mathfrak{S}_e\,\mathfrak{S}_v=\mathfrak{S}_v$ is a single term, and it accounts for everything in degree 0; yet the double product $\mathfrak{S}_e(x;y)\,\mathfrak{S}_v(x;z)$ has exactly $\big|[e,v]_L\big|$ terms, so all but one of them lie in positive degree. No estimate of the degree-0 count was needed to see this, and none could have helped: a bound extracted from degree 0 alone would have returned 1.

This is why no such bound is available for the ordinary product. Its support lies entirely in degree $\ell(u)+\ell(v)$, so counting it means counting exactly the coefficients that resist counting; there is no second family to appeal to. This is also why the question Remmel and Whitney raised is not settled here: their $|C(\lambda*\mu)|$ counts tableaux for a product of two Schur functions, with multiplicity and in a single degree, and the mechanism just described is unavailable to it. The double product supplies one. It contains the whole independent family $\{\mathfrak{S}_u\,\mathfrak{S}_{v''} : v''\leq_L v\}$ within a single support, spread across the ranks of $[e,v]_L$ by Theorem 5.16, and it is the size of that family—a purely order-theoretic quantity, available for free—that the argument converts into a term count. The coefficient variables are not an inconvenience of the double case but the source of the count.

Corollary 5.20 (Worst-case output-polynomiality, for each fixed u)**.** *Fix n and regard* **schubmult** *as an algorithm* **schubmult**$_n$ *on inputs $(u,v)\in S_n\times S_n$. For $u\in S_n$ let*

$$W(u,n):=\max_{v\in S_n}T(u,v),\qquad N^*(u,n):=\max_{v\in S_n}N(u,v),$$

where $T(u,v)$ is the bound of Theorem 5.11 and $N(u,v)$ is the number of terms of $\mathfrak{S}_u(x;y)\,\mathfrak{S}_v(x;z)$ in the Schubert basis. Then for all sufficiently large n and every $u\in S_n$,

$$n!\ \leq\ N^*(u,n)\ \leq\ W(u,n)\ \leq\ (n!)^3,$$

and consequently

$$W(u,n)\ \leq\ N^*(u,n)^3\qquad \textit{for all } u\in S_n.$$

Proof. The bound $N^*(u,n)\geq n!$ is Proposition 5.18 at $v=w_0(n)$, where $[e,w_0]_L=S_n$; that proposition is uniform in u, so the single input (u,w_0) witnesses it for whichever u is fixed. The inequality $N^*(u,n)\leq W(u,n)$ is that the algorithm must emit each of the $N(u,v)$ coefficients.

For the upper bound, $P_uV\leq(2n-1)!!\,n!=(2n)!/2^n=\binom{2n}{n}(n!)^2/2^n\leq 2^n(n!)^2$. With $L\leq n$, $B\Delta\leq 4^n$ and $E_n=2^{\Theta(\log^2 n)}$ this gives $T(u,v)\leq n\,8^n(n!)^2E_n$ for every pair, and $n\,8^nE_n=2^{3n+\Theta(\log^2 n)}\leq n!$ once $\log_2 n>3+\log_2 e+o(1)$, by Stirling. Equivalently, at the level of exponents,

$$\log_2\big((n!)^3\big)-f(n)=n\log_2 n-(3+\log_2 e)\,n-\Theta(\log^2 n)\longrightarrow+\infty,$$

so the third factor of $n!$ is not consumed and the containment has slack $(n!)^{1-o(1)}$. Combining, $W(u,n) \le (n!)^3 \le N^*(u,n)^3$. $\square$

Remark 5.21 (Scope of the preceding corollary)*.* Since $N^*(u,n)$ is the largest number of coefficients that must be written, it is a lower bound on the worst-case cost, over v, of any algorithm that writes these products; Corollary 5.20 therefore bounds the cost of $\mathbf{schubmult}_n$ by a fixed polynomial in the size of its output, in the worst case over v alone, with u held fixed and arbitrary. Taking the maximum over u as well recovers $W(n) \le N^*(n)^3$.

The pointwise notion of an output-polynomial algorithm, or one running in polynomial total time—running time bounded by a polynomial in the input size plus the output size, on every instance—is due to Johnson, Yannakakis, and Papadimitriou [6]. Corollary 5.20 is a worst-case relaxation of it: the maximum is taken over v before the comparison, so a large output at one v funds a large running time at every v. It does not assert the pointwise statement $T(u,v) \le N(u,v)^3$, which from the same two bounds would follow only with a correction,

$$T(u,v) \;\le\; N(u,v)^3 \cdot \Big(\frac{n!}{\big|[e,v]_L\big|}\Big)^3,$$

not controlled: since $[e,v]_L \subseteq S_n$, the value $\big|[e,v]_L\big| = n!$ forces $v = w_0$, whereas for v a simple reflection $\big|[e,v]_L\big| = 2$ while $\Lambda(u,v)$ need not fall with it. Deciding the pointwise question would require replacing $\Lambda(u,v)$, which counts an entire order ideal, by a count of the permutations the algorithm actually visits; Proposition 5.18 bounds $N(u,v)$ only from below.

What the corollary does establish is that the superpolynomial running time of Theorem 5.11 is not an artifact of loose accounting. Since $\log_2 W(u,n) \le 3\log_2 N^*(u,n)$, the exponent is within a factor of three of the information-theoretic floor, for every u separately; and since the input (u,v) occupies $O(n\log n)$ bits while the output may have $n! = 2^{\Theta(n\log n)}$ terms, no algorithm writing these products in the Schubert basis can run in time polynomial in the input length.

Remark 5.22 (Comparison with prior complexity results)*.* The complexity of a single structure constant has been studied more than that of an entire product. For Schur polynomials the coefficients $c^{\nu}_{\lambda\mu}$ are the Littlewood–Richardson coefficients, and computing one of them is $\#P$-complete [10]. Since these coefficients can be exponentially large, a single one has polynomial bit-length yet cannot be evaluated in time polynomial in the binary sizes of the input and output unless FP $= \#P$; the same then holds for the full product $s_\lambda s_\mu = \sum_\nu c^{\nu}_{\lambda\mu}\, s_\nu$. The Littlewood–Richardson rule instead enumerates skew tableaux, of which there are $\sum_\nu c^{\nu}_{\lambda\mu}$, and its running time is polynomial in that number, namely in the coefficients written in unary. Deciding whether a given coefficient is positive is, by contrast, in P, through the saturation theorem [7] and the reduction to hive-polytope nonemptiness [4].

For Schubert polynomials the coefficients c^{w}_{uv} generalize the Littlewood–Richardson coefficients, recovered in the Grassmannian case, so computing one is $\#P$-hard; whether they lie in $\#P$ is open, no combinatorial rule being known. The vanishing problem $\{c^{w}_{uv} = 0\}$ was recently placed in the polynomial hierarchy, in *coAM* assuming the generalized Riemann hypothesis, hence in Σ^p_2 [11].

Since a single coefficient is already $\#P$-hard, neither the bound of Theorem 5.11 nor that of Corollary 5.13 is polynomial in the binary output size, and the two are compared instead per combinatorial witness. Each executed iteration of **schubmult** processes one matched (cover, triple) pair. Corollary 5.13 performs $O(1)$ integer operations at each such pair, so in the single case the number of ring operations is proportional to the number of witnesses. Theorem 5.11 performs $E_n = n^{\Theta(\log n)}$ operations at each pair (Proposition 5.10), a quasi-polynomial factor—superpolynomial but subexponential in n—incurred by rebuilding the factorial elementary symmetric weight; the single specialization removes it because that weight collapses to a sign.

Two clarifications are worth making about the Grassmannian bounds of Section 6, since polynomiality there is easy to misread. First, the polynomiality is in the output, not in the input. Corollary 6.5 bounds the cost of the entire product by a degree-four polynomial in G_k, and G_k bounds the number of coefficients returned. Some such reading is forced: the input λ,μ occupies $O(k\log n)$ bits in binary while the output may carry $\Theta(n^k)$ terms, exponentially many in the input size, so no algorithm for the full product runs in time polynomial in the input, for reasons having nothing to do with $\#P$-hardness. Measured against the output the comparison with the classical rule is favorable: the Littlewood–Richardson rule enumerates $\sum_\nu c^{\nu}_{\lambda\mu}$ tableaux, which is the coefficient mass in unary and can be exponentially larger than the number of nonzero terms, whereas the bound here is polynomial in the number of terms, with each coefficient carried in factored form. We do not claim asymptotic optimality. Writing the output costs $\Omega(G_k)$ operations in the worst case. In the single case G_k overcounts the support of one product by at most the factor $2k(n-k)+1$ recording the available degrees, that support lying in the single degree $|\lambda|+|\mu|$; in the factorial case no such collapse is available, since by Theorem 5.16 the support is a union of ordinary supports indexed by the left weak order ideal below v and so meets every degree between ℓ_u and $\ell_u+\ell_v$. Either way the information-theoretic floor is linear in G_k up to a polynomial factor, and the gap to the degree-four bound is at most G_k^3. Second, the resulting statement that for fixed k the count is polynomial in n is not itself new, and is in fact weaker than what is known for a single coefficient: the parts of the partitions involved are bounded by $2(n-k)$, so polynomiality in n is polynomiality in the unary size of the input. For a fixed number of rows the hive polytope of [7] has fixed dimension, and Barvinok's lattice-point algorithm [1] evaluates a single Littlewood–Richardson coefficient in time polynomial in the binary size of the partitions; this is stated explicitly in [5]. What Section 6 contributes is therefore not fixed-rank polynomiality but the explicit exponent $4(2n-k)H_2\big(k/(2n-k)\big)$, valid uniformly in k and bounding the cost of the whole expansion by $2^{O(n)}$ rather than $n^{\Theta(n)}$.

6. The Grassmannian case: Schur and factorial Schur multiplication

When both factors are Grassmannian the product specializes to Schur, respectively factorial Schur, multiplication, and the doubled-staircase support of Lemma 5.3 collapses onto a single box. This tightens the counts of Definition 5.8 and, more consequentially, lowers the leading exponent of the running time from $\Theta(n\log n)$ to $4(2n-k)H_2\big(k/(2n-k)\big)$, with H_2 the binary entropy function: a genuine function of the two box dimensions k and $n-k$. The case to keep in mind is the balanced one $k=n/2$, where the relevant count is the central quantity

$$G_{n/2}=\binom{3n/2}{n/2}=(27/4)^{\,n/2+o(n)}=2.5980\ldots^{\,n(1+o(1))}$$

and the bound G_k^4 reads $(27/4)^{2n+o(n)}=(729/16)^{\,n+o(n)}$, an exponent of $2n\log_2(27/4)=5.5098\ldots n$. This is essentially the worst case: the maximum of the exponent over all k is $8n\log_2\varphi=5.5539\ldots n$ for φ the golden ratio, attained at $k\approx 0.553\,n$, less than one percent above the balanced value. The degenerate dimensions are the ones that gain, the exponent vanishing as $k\to 0$ or $k\to n$, and for k held fixed the cost drops to $O(n^{4k})$.

Definition 6.1 (k-Grassmannian permutation)**.** A permutation $w\in S_n$ is *k-Grassmannian* if its unique descent lies at position k, that is $w(1)<\cdots<w(k)$ and $w(k+1)<\cdots<w(n)$ with $w(k)>w(k+1)$. To such a w we associate the partition $\lambda(w)$ with parts

$$\lambda(w)_i=w(k-i+1)-(k-i+1),\qquad 1\le i\le k,$$

which has at most k parts, each at most $n-k$; that is, $\lambda(w)$ is contained in the $k\times(n-k)$ box. This assignment is a bijection between k-Grassmannian permutations in S_n and partitions in that box.

For a k-Grassmannian u the single Schubert polynomial $\mathfrak{S}_u(x)$ equals the Schur polynomial $s_{\lambda(u)}(x_1,\ldots,x_k)$, and the double Schubert polynomial $\mathfrak{S}_u(x;z)$ equals the factorial Schur polynomial $s_{\lambda(u)}(x;z)$. When u and v are both k-Grassmannian the structure constants of $\mathfrak{S}_u\cdot\mathfrak{S}_v$ are therefore the Littlewood–Richardson coefficients $c^{\nu}_{\lambda(u)\lambda(v)}$ in the single case [3] and the factorial (Littlewood–Richardson) polynomials in the double case [9, 15].

Lemma 6.2 (Grassmannian output support)**.** *Let $u,v\in S_n$ be k-Grassmannian. Then $\mu(v^{-1})$ is k-Grassmannian, and every permutation occurring in the product $\mathfrak{S}_u\cdot\mathfrak{S}_v$, or in any intermediate product formed during its computation by* **schubmult***, is k-Grassmannian with associated partition contained in the $k\times 2(n-k)$ box. Both the u-side permutations and the v-side labels are of this form. The number of such permutations is at most*

$$G_k:=\binom{2n-k}{k},$$

the number of partitions in the $k\times 2(n-k)$ box.

Proof. That $\mu(v^{-1})$ is k-Grassmannian, and that every intermediate u-side permutation and v-side label stays k-Grassmannian, is the structural property of the Grassmannian shape, which we take as given, exactly as the doubled-staircase support was taken as given in Lemma 5.3. It remains to bound the associated partitions. Write $\lambda := \lambda(u)$ and $\mu := \lambda(\mu(v^{-1}))$, both contained in the $k \times (n-k)$ box, so $\lambda_1, \mu_1 \leq n-k$. Any partition ν arising in the product or a partial product satisfies $\nu \supseteq \lambda$, has at most k parts, and obeys $\nu_1 \leq \lambda_1 + \mu_1 \leq 2(n-k)$; hence ν is contained in the $k \times 2(n-k)$ box. The number of partitions in a $k \times 2(n-k)$ box is $\binom{k+2(n-k)}{k} = \binom{2n-k}{k} = G_k$. For fixed k this is $\binom{2n-k}{k} = \Theta(n^k)$, a polynomial of degree k in n; uniformly in k it obeys

$$G_k \;\leq\; \sum_{j\geq 0} \binom{2n-j}{j} \;=\; F_{2n+1} \;=\; \Theta(\varphi^{2n}),$$

the $(2n{+}1)$-st Fibonacci number, with $\varphi = (1+\sqrt{5})/2$ and $\varphi^2 = (3+\sqrt{5})/2 < 2.619$; hence $G_k = O(\varphi^{2n}) = O(2.619^{\,n})$. Jointly in n and k, Stirling's approximation for binomial coefficients gives the closed form

$$\log_2 G_k = (2n-k)\, H_2\!\left(\frac{k}{2n-k}\right) + O(\log n), \qquad H_2(x) := -x\log_2 x - (1-x)\log_2(1-x),$$

with H_2 the binary entropy function; since $0 \leq H_2 \leq 1$ this recovers $\log_2 G_k \leq 2n-k$. The bound $\binom{m}{k} \leq (em/k)^k$ with $m = 2n-k$ gives the sharper reading

$$\log_2 G_k \;\leq\; k\log_2 \frac{2n}{k} + k\log_2 e,$$

which for fixed k is $k\log_2 n + O(k)$, matching $G_k = \Theta(n^k)$, and which shrinks to $O(\log n)$ as $k \to 0$; the symmetric bound $\binom{2n-k}{k} \leq (k+1)^{2(n-k)}$ gives the same as $k \to n$. Uniformly in k the entropy expression is maximized at $2n\log_2\varphi$, attained near $k \approx 0.553\,n$, in agreement with the Fibonacci bound above. The running time is governed by the two box dimensions k and $n-k$, never by n alone. Finally, every intermediate shape lies in the same $k \times 2(n-k)$ box, so a single Pieri step produces at most G_k distinct covers; hence the per-step branchings obey $B, \Delta \leq G_k$, and the number of layers is at most $n-k$ (the code of v^{-1} has at most $n-k$ nonzero parts). ☐

Theorem 6.3 (Complexity in the Grassmannian case)**.** *Let $u, v \in S_n$ be k-Grassmannian and set $G_k = \binom{2n-k}{k}$. With $L \leq n-k$ layers and per-step branchings $B, \Delta \leq G_k$ (Lemma 6.2), the number of coefficient-ring operations performed by* **schubmult**(u,v) *is*

$$O\big(\, n^2\,\Phi_v \;+\; L\cdot G_k \cdot B \cdot n^3 \;+\; L \cdot G_k^2 \cdot B \cdot \Delta \cdot E_n\,\big)\,,$$

with E_n as in Definition 5.8. Writing this bound as $2^{f_{\mathrm{Gr}}(n,k)+O(\log n)}$, the worst case is bounded by the explicit function of n and k

$$f_{\mathrm{Gr}}(n,k) = 4(2n-k)\, H_2\!\left(\frac{k}{2n-k}\right) + \Theta(\log^2 n), \qquad H_2(x) = -x\log_2 x - (1-x)\log_2(1-x),$$

with H_2 the binary entropy function (Lemma 6.2). Equivalently the bound is $O(n\, G_k^4\, E_n)$.

Proof. The three-part accounting of the proof of Theorem 5.11 applies verbatim; only the two counts and the branchings change. By Lemma 6.2 both the u-side permutations and the v-side labels are k-Grassmannian with partition in the $k \times 2(n-k)$ box, so the u-side count P_u and the v-side label count V are each replaced by G_k: the u-side count G_k confines the distinct u-permutations, and the v-side count G_k replaces the general bound $V \leq n!$, because the labels are now k-Grassmannian rather than ranging over all of S_n. The hash therefore holds at most G_k^2 keys at once. The live **pieri** enumeration runs once per distinct u-permutation per layer, at most $L \cdot G_k$ times, each $O(n^3 \cdot B)$; the matched-pair work is over $W \leq L \cdot G_k^2 \cdot B \cdot \Delta$ pairs, each building one factorial elementary symmetric polynomial at cost E_n; and the v-side precomputation is $O(n^2 \Phi_v)$. Summing gives the stated bound, whose worst-case term is the third.

For the worst case take base-2 logarithms of the third summand. Every shape lies in the $k \times 2(n-k)$ box, so a single Pieri step produces at most G_k covers and $B, \Delta \leq G_k$; the number of layers L equals the number of nonzero parts of the code of v^{-1}, at most $n-k$; and $\log_2 E_n = \Theta(\log^2 n)$ (Proposition 5.10). Substituting,

$$\log_2 T \leq \log_2(n-k) + 4\log_2 G_k + \Theta(\log^2 n) = 4\log_2 G_k + \Theta(\log^2 n) + O(\log n).$$

By Lemma 6.2, $\log_2 G_k = (2n-k)\,H_2\!\left(\frac{k}{2n-k}\right) + O(\log n)$ with H_2 the binary entropy function, so

$$f_{\mathrm{Gr}}(n,k) = 4(2n-k)\,H_2\!\left(\frac{k}{2n-k}\right) + \Theta(\log^2 n) \ \le\ 8n\log_2\varphi + \Theta(\log^2 n).$$

The exponent is controlled by the box dimensions k and $n-k$. At $k = n/2$ one has $k/(2n-k) = 1/3$ and $H_2(1/3) = \log_2 3 - \frac{2}{3}$, so $f_{\mathrm{Gr}}(n,n/2) = 6n\,H_2(1/3) + \Theta(\log^2 n) = 2n\log_2(27/4) + \Theta(\log^2 n) = 5.5098\ldots n + \Theta(\log^2 n)$. The exponent collapses to $\Theta(\log^2 n)$ at the extremes $k \to 0$ and $k \to n$, and its maximum over k is $8n\log_2\varphi = 5.5539\ldots n$, attained near $k \approx 0.553\,n$, matching the Fibonacci bound $G_k = O(\varphi^{2n})$ of Lemma 6.2; the balanced value sits within one percent of it. $\square$

Remark 6.4. The general bound of Theorem 5.11 has dominant exponent $\log_2((2n)!) - n = \Theta(n\log n)$, contributed by the $P_u \le (2n-1)!!$ distinct u-side permutations and the $V \le n!$ distinct v-side labels. In the Grassmannian case both counts, and every per-step branching, are confined to the $k\times 2(n-k)$ box and hence bounded by $G_k = \binom{2n-k}{k}$ (Lemma 6.2), so the whole exponent collapses to $4\log_2 G_k = 4(2n-k)H_2\!\left(\frac{k}{2n-k}\right) + O(\log n)$. This is a genuine function of both box dimensions, and its constants are what the collapse consists of. At the balanced dimension $k = n/2$ it is $4\log_2\binom{3n/2}{n/2} = 6n\,H_2(1/3) = 2n\log_2(27/4) = 5.5098\ldots n$, using $H_2(1/3) = \log_2 3 - \frac{2}{3}$, so the cost is $(729/16)^{\,n+o(n)}$; and this is within one percent of its maximum over k, namely $8n\log_2\varphi \approx 5.554\,n$ attained near $k \approx 0.553\,n$; it shrinks to $\Theta(\log n)$ only as $k \to 0$ or $k \to n$, mirroring the shrinking of $\mathrm{Gr}(k,n)$ at the extremes, so the leading exponent $\Theta(n\log n)$ of Theorem 5.11 never appears. The improvement over the general algorithm is a factor $n^{\Theta(n)} = 2^{\Theta(n\log n)}$ in the worst-case bounds. Equivalently the cost is polynomial, of degree at most four, in the number G_k of Grassmannian outputs, and for k held fixed $G_k = \Theta(n^k)$ gives $O(n^{4k})$; Remark 5.22 discusses what such a bound does and does not say.

Corollary 6.5 (Schur multiplication and Littlewood–Richardson coefficients)**.** *Let $u, v \in S_n$ be k-Grassmannian. Computing the structure constants of $\mathfrak{S}_u(x)\,\mathfrak{S}_v(x)$—the Littlewood–Richardson coefficients $c^{\nu}_{\lambda(u)\lambda(v)}$—by the single-variable specialization of* **schubmult** *takes*

$$O\big(\min(n-1,\ell_v)\cdot G_k^2\cdot B\cdot\Delta\big)$$

integer operations, with $L \le n-k$ layers and $B, \Delta \le G_k$. Writing this as $2^{g_{\mathrm{Gr}}(n,k)+O(\log n)}$, the worst case is bounded by the explicit function of n and k

$$g_{\mathrm{Gr}}(n,k) = 4(2n-k)\,H_2\!\left(\frac{k}{2n-k}\right), \qquad H_2(x) = -x\log_2 x - (1-x)\log_2(1-x),$$

with H_2 the binary entropy function; this is the exponent of Theorem 6.3 with the term $\Theta(\log^2 n)$ deleted.

Proof. As in Corollary 5.13, the single-variable weight attached to a matched (cover, triple) pair is the integer sign $s \in \{+1,-1\}$ rather than a factorial elementary symmetric polynomial, so each executed iteration performs $O(1)$ integer operations and E_n is replaced by $O(1)$ in the third summand. The three-part bound of Theorem 6.3 then reads $O(n^2\Phi_v + L\cdot G_k\cdot B\cdot n^3 + L\cdot G_k^2\cdot B\cdot\Delta)$, and the substitutions $L \le n-k$, $B, \Delta \le G_k$ (Lemma 6.2), $\log_2 G_k = (2n-k)\,H_2\!\left(\frac{k}{2n-k}\right) + O(\log n)$ give $g_{\mathrm{Gr}}(n,k) = 4(2n-k)\,H_2\!\left(\frac{k}{2n-k}\right)$. In particular the operation count is polynomial in the number G_k of Grassmannian outputs, of degree at most four, consistent with Remark 5.22: an individual coefficient $c^{\nu}_{\lambda(u)\lambda(v)}$ may be exponentially large in binary, but the algorithm's arithmetic is counted per output rather than per bit. $\square$

References

[1] Barvinok, A., and Woods, K. Short rational generating functions for lattice point problems. *J. Amer. Math. Soc. 16*, 4 (2003), 957–979.

[2] Bergeron, N., and Sottile, F. Skew Schubert functions and the Pieri formula for flag manifolds. *Trans. Amer. Math. Soc. 354*, 2 (2002), 651–673.

[3] Buch, A. S. A Littlewood–Richardson rule for the K-theory of Grassmannians. *Acta Mathematica 189*, 1 (2002), 37–78.

[4] Bürgisser, P., and Ikenmeyer, C. Deciding positivity of Littlewood–Richardson coefficients. *SIAM Journal on Discrete Mathematics 27*, 4 (2013), 1639–1681.

[5] De Loera, J. A., and McAllister, T. B. On the computation of Clebsch–Gordan coefficients and the dilation effect. *Experiment. Math. 15*, 1 (2006), 7–19.

[6] Johnson, D. S., Yannakakis, M., and Papadimitriou, C. H. On generating all maximal independent sets. *Information Processing Letters 27*, 3 (1988), 119–123.

[7] Knutson, A., and Tao, T. The honeycomb model of $GL_n(\mathbb{C})$ tensor products I: Proof of the saturation conjecture. *Journal of the American Mathematical Society 12*, 4 (1999), 1055–1090.

[8] Macdonald, I. G. *Notes on Schubert Polynomials.* Université Du Québec à Montréal, Dép. de mathématiques et d'informatique, 1991.

[9] Molev, A., and Sagan, B. A Littlewood-Richardson rule for factorial Schur functions. *Trans. Amer. Math. Soc. 351*, 11 (1999), 4429–4443.

[10] Narayanan, H. On the complexity of computing Kostka numbers and Littlewood-Richardson coefficients. *J. Algebraic Combin. 24*, 3 (2006), 347–354.

[11] Pak, I., and Robichaux, C. Vanishing of Schubert coefficients. *arXiv preprint arXiv:2412.02064* (2024). with an appendix joint with David E. Speyer.

[12] Remmel, J. B., and Whitney, R. Multiplying Schur functions. *J. Algorithms 5* (1984), 471–487.

[13] Samuel, M. J. A Pieri formula for multiplying quantum double Schubert polynomials by consolidated pairs of factorial quantum elementary symmetric polynomials. In preparation.

[14] Samuel, M. J. A Pieri formula for quantum double Schubert polynomials. In preparation.

[15] Samuel, M. J. A Molev-Sagan type formula for double Schubert polynomials. *J. Pure Appl. Algebra 228*, 7 (2024), 107636.

[16] Samuel, M. J. `schubmult`: a Python package for multiplying Schubert polynomials. Available from the Python Package Index, `https://pypi.org/project/schubmult/`, with sources at `https://github.com/matthematics/schubmult`, 2026.